

DES311 Project
Level design

Nicole Meenan
2102039

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Pre-production

Game concept – Alicia in Fantasyland

Inspired by Alice in wonderland, a game where you play as Alicia, a character who can consume collectables that will cause her to shrink and grow. She will use her growth to overcome obstacles and reach high points and shrink to fit through passageways and avoid obstacles. This causes the player to strategize to tackle challenges. I want the environment to feel whimsical and dream-like, such as the original.







Moodboard

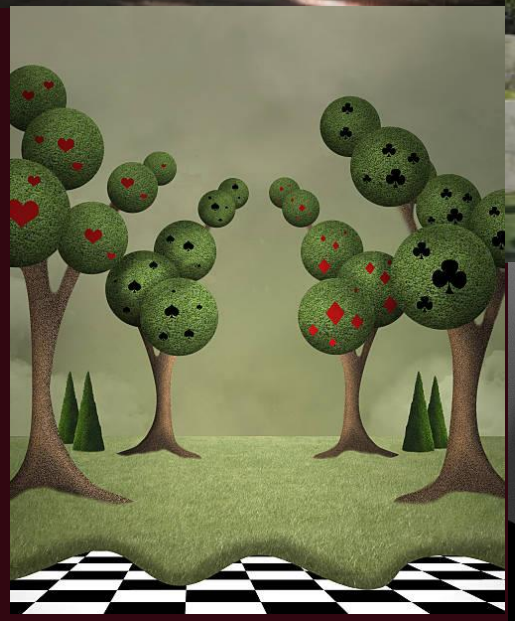


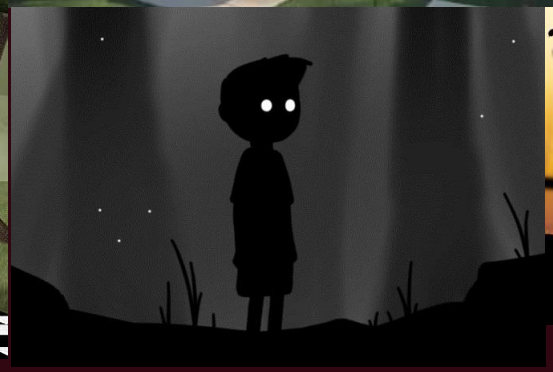


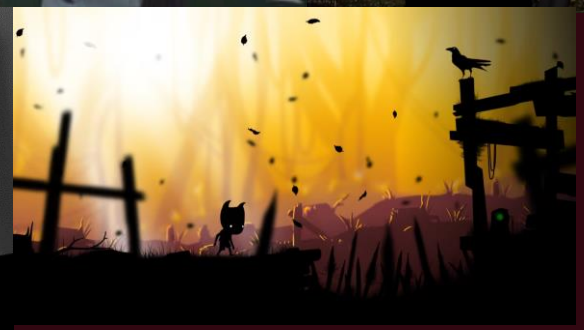












Lighting & Colour

















Aims

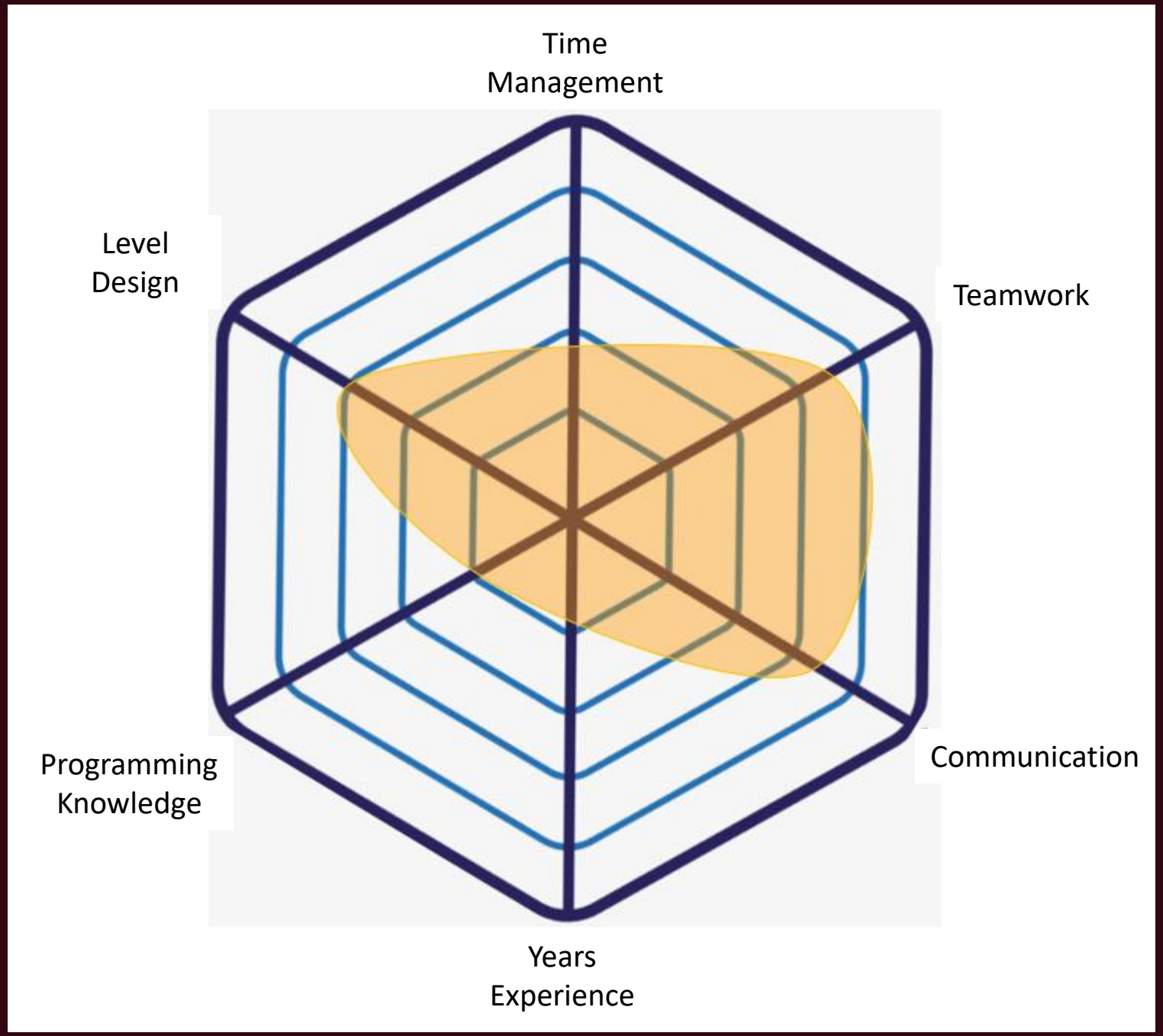
- Develop my level design and programming knowledge in preparation for industry.
- Create mechanics that feel smooth and polished.
- Achieve level design that flows well with the mechanic.
- Conduct testing often and improve UX.
- Study level design with use of academic sources.
- Implement game design principles into my level.

Mechanics

- The player will be able to navigate through a 3D space
- They will collect 2 variations of collectables that will allow them to shrink and grow.
- Their jump and speed will differ accordingly to their size.

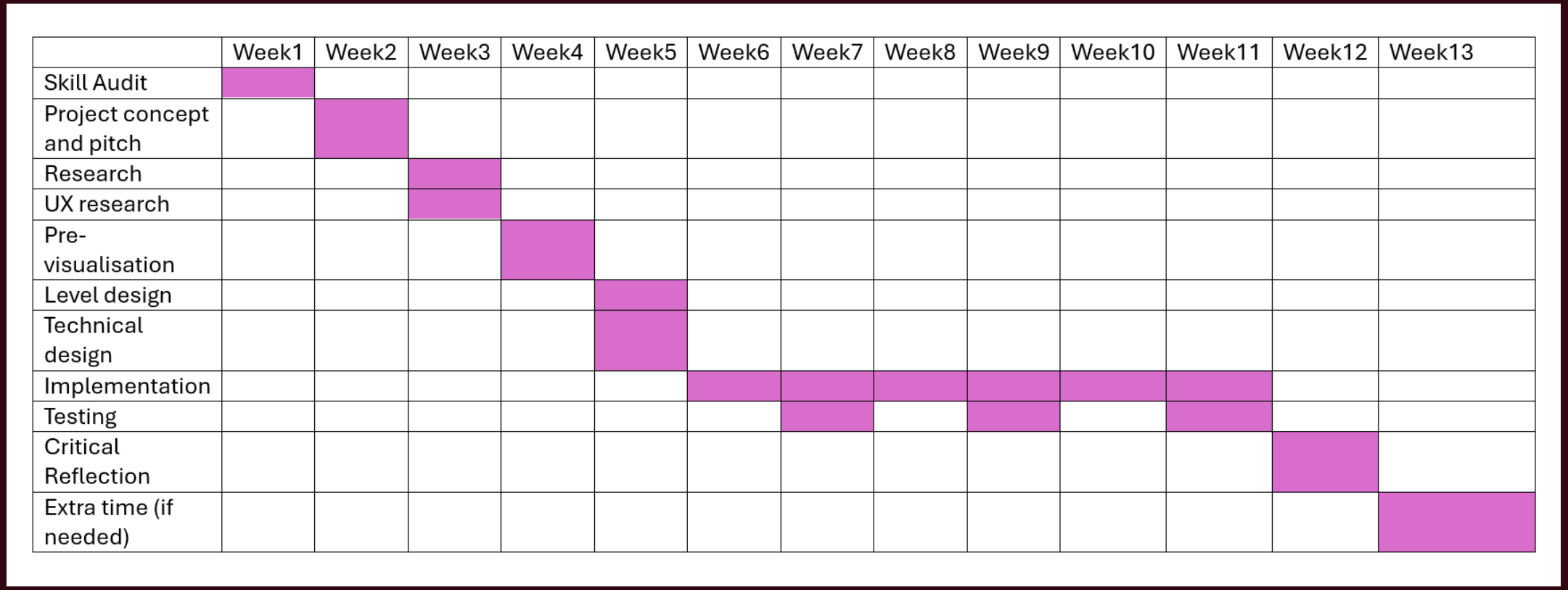
Skills Diamond

In reference to my skills diagram, I noticed there was a gap within my programming knowledge as this is a large, complex skill to learn. My only knowledge of anything game design related has been through my education at Abertay, no background knowledge. This has made my skill set broad and well-rounded. I particularly took interest in level design and programming in C# and blueprints. This project will further develop my level design knowledge, as well as programming knowledge using Unreal Engine's blueprints system. A stretch goal of mine is to begin learning C++.



Scheduling

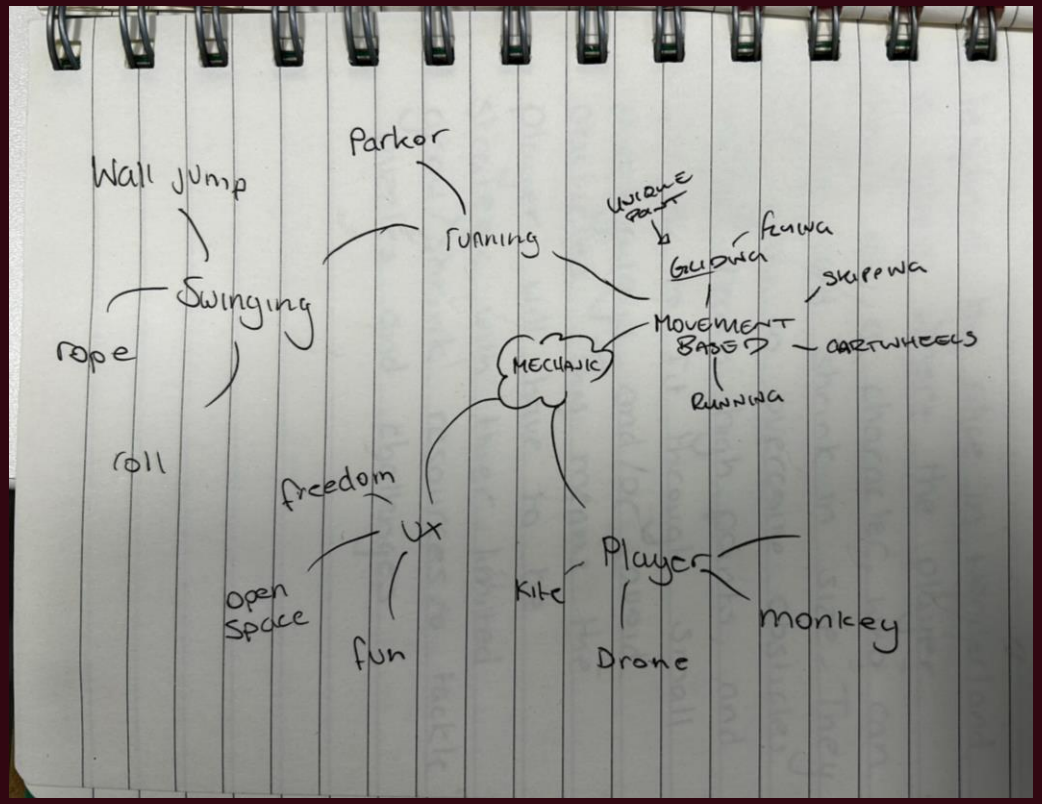
I used a gantt chart to plan my time appropriately for the scope of the project.



	Week1	Week2	Week3	Week4	Week5	Week6	Week7	Week8	Week9	Week10	Week11	Week12	Week13
Skill Audit													
Project concept and pitch													
Research													
UX research													
Pre-visualisation													
Level design													
Technical design													
Implementation													
Testing													
Critical Reflection													
Extra time (if needed)													

UX Research

Before thinking about anything else, I initially considered my player experience goals. I wanted the player to feel a sense of freedom without creating something huge like a sandbox game. I also wanted the game to be movement based and for the player to feel satisfied when navigating through the level, nothing too frustrating or challenging. After brainstorming some ideas, I thought this would be most realistic for my skillset whilst meeting these specifications for user experience. My first stage mind map seen on the right.



Pre-Production – Research

Effective design of video games

When planning this project and deciding on a mechanic focus, it was important that the player didn't feel overly challenged and that the level flowed cohesively alongside the mechanic. In the early phases, inspiration was taken from *'Alice: Madness Returns'* (Spicy Horse, 2011). As this shared similar qualities being a single player action-adventure game under the same theme.

In design, it is important to maintain consistency as this subconsciously helps players make connections between principles, making progression much smoother. Linear progression thrives when players follow repetitive skill or memorized facts (Dondlinger, 2007).

The way that games universally communicate this is through shape, colour, lighting and audio. Reiteration of these elements when provoking a player to behave in an intended way, is extremely effective.

Level Design – Proportion.

This is a tricky area particularly for my mechanic as the player will be varying between 3 different scales. Designing a level around this while still having it feel unified is a challenge.

Proportion doesn’t necessarily affect this project directly since the nature of the core mechanic and the fantasy environment, things not exactly feeling proportional can be justified.

Gameplay mechanics heavily determine what size the level and props should be to create harmony. In the case of this project, I can use the surface area of objects as well as their nooks to benefit the player. With the ability to shrink and grow this expands the horizon of possibilities within a single prop or area.

Level design - Weenies



The layout and flow of the Disneyland theme parks is something that is referenced often when discussing the topic of improvement within level design.

The imagineers utilised the use of architectural landmarks - most famously the castle – to guide guests and aid their direction. They called these “Weenies”. Weenies do not have to be huge landmarks; it can be anything that stands or catches the eye (Rogers, 2014).

This can be seen in video games such as *The Last of Us* (Naughty Dog, 2013) where there is clear guidance with use of distant landmarks and camera position, where the player is intended to go.



In this project, the “weenies” are doors to the following room, prompting the player to find the collectable that allows them to access it. These will be the first thing visible to the player when entering a new area.

Level Design – Keeping the player engaged

Things that capture players attention:

- Sudden appearances
- Colour changes
- Looming motion
- Size changes

(Steamworks Development, 2016)

The mechanic that my level is designed around will incorporat lighting colour changes to show progression within the level, size changes of the player and the illusion of looming motion due to these size changes.

Adding plenty of variety can avoid boredom. Changing the direction the player moves can be enough to keep them alert, elevation particularly. The differentiation of colour, textures can make all the difference too (Rodgers, 2014). Time of day particularly is a simple tool that can completely transform an environment.



Firewatch (Campo Santo, 2016)

Human minds can focus attention on things and almost forgetting the outside world. In complete immersion where thoughts lessen, and time passes quickly. This is what most experience when playing a truly well-designed game. Therefore, a game designer must be able to study this flow and iterate it in their work Considering this, the 4 design techniques that achieve this basis are:

- Clear goals
- No distractions
- Direct feedback
- Continuously challenging

(Schell, 2020).

The designer must balance these elements without reaching any extremes as this would have the opposite effect than intended.

Player Psychology

The psychology of rewards.

Rewards can sometimes have the opposite effect than intended. Players become infatuated with the feeling chasing a reward gives them that they can often avoid main intended gameplay quests that feel more tedious or risky. The developers or the game *Outer Wild* (Mobius Digital, 2019) deliberately avoided giving players specific goals about where to go or what to achieve so that players would move through the game with their curiosity of the unknown driving them. I will design my level with no standout “reward” just linear progression. So how will I keep the player engaged? Giving them small paybacks for exploration out with the intended path.

Intrinsic vs Extrinsic Motivation

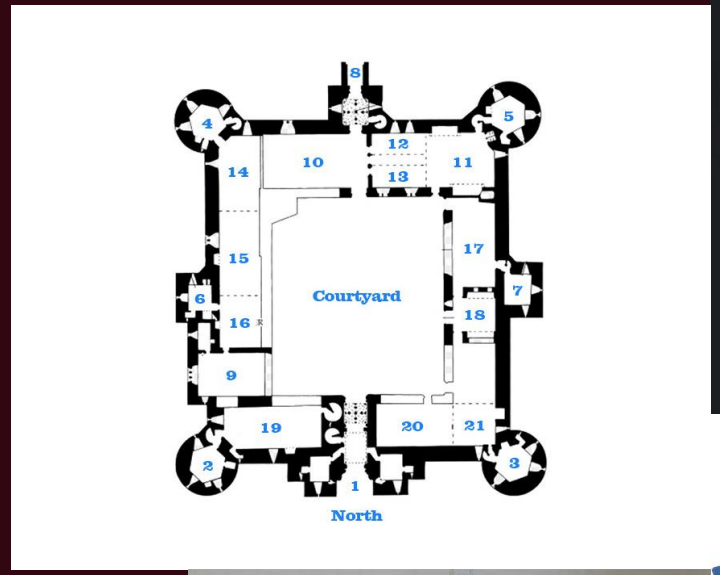
The difference between intrinsic and extrinsic motivation is doing something for yourself, or doing something for an external factor, such as going to work for money. Successful games are designed with a complete balance of both. Or even better, can convince players they are acting intrinsically when it's the contrary. There is a fine line between the two and – without getting too philosophical - people often can't distinct between them themselves, this is a means that can be cleverly utilised by designers.

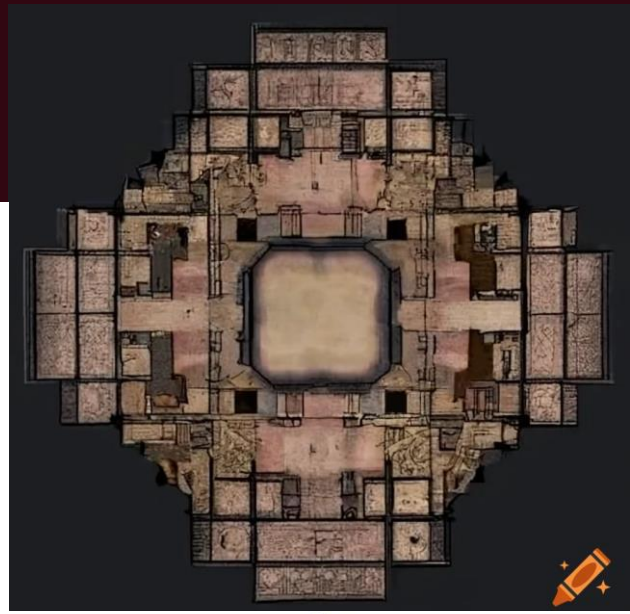
It is found that work conducted by those with extrinsic motivations is of lower quality and shorter lived than those who are intrinsically motivated (Deci, 1971).

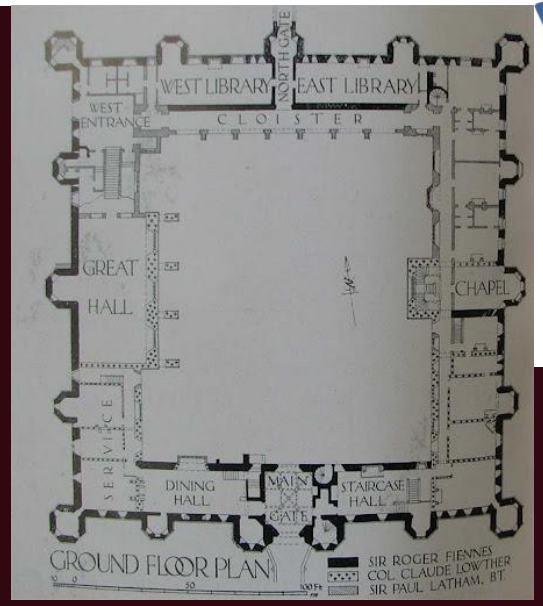

Production

Level Design

Level design is inspired by the castle in the movie combined with linear platforming elements. I looked at castle floor plans and opted for something simple and symmetrical often seen in fantasy settings. The player will move around the loop of hallways and rooms to search for a way to the central courtyard where they will burrow through a rabbit hole and escape.



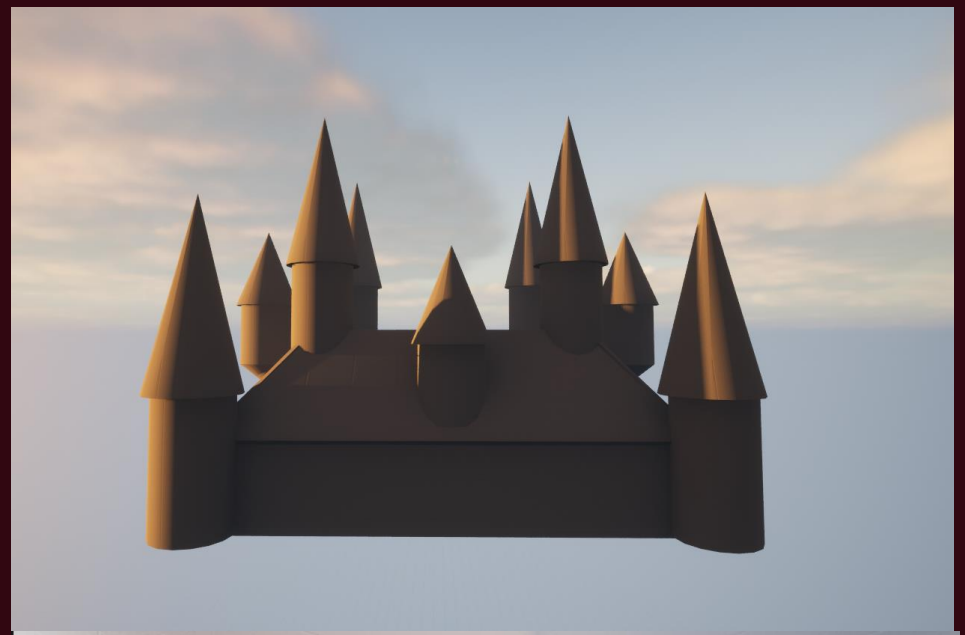


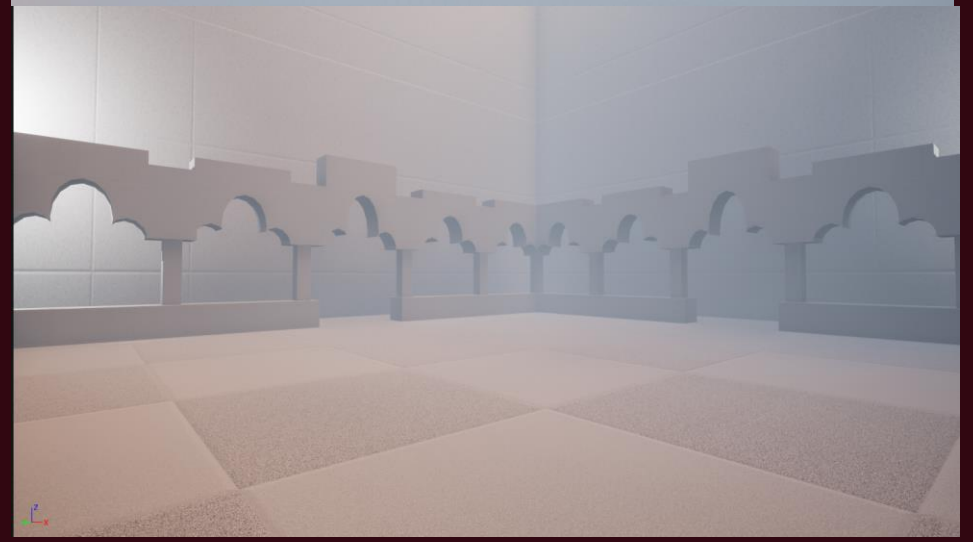




Level design – First iteration

I created a symmetrical castle using my first level design images as reference. I blocked off two levels and added spiral stairs to the turrets. The courtyard was open, and natural light would come in from above. The environment was very large in scale of the player.



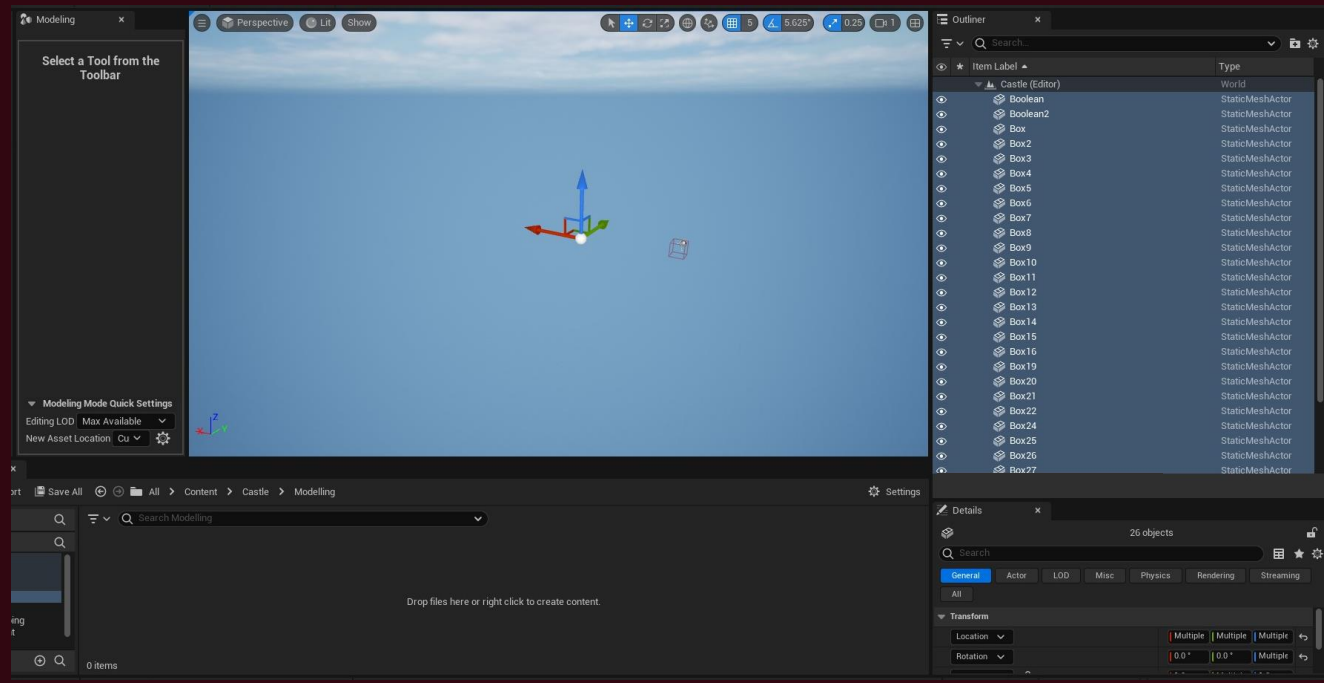


Testing (V1)

- The setting communicated the concept I am trying to portray.
- Walking through this environment felt repetitive and unnatural, it didn’t spark ideas for how to create progression.
- The scale is too large for the player so it would become difficult to incorporate the mechanic through this space.
- I felt as though this is restricting the player and there is not enough diversity between rooms. I aimed to avoid repetition, so I redesigned the level after this initial version.

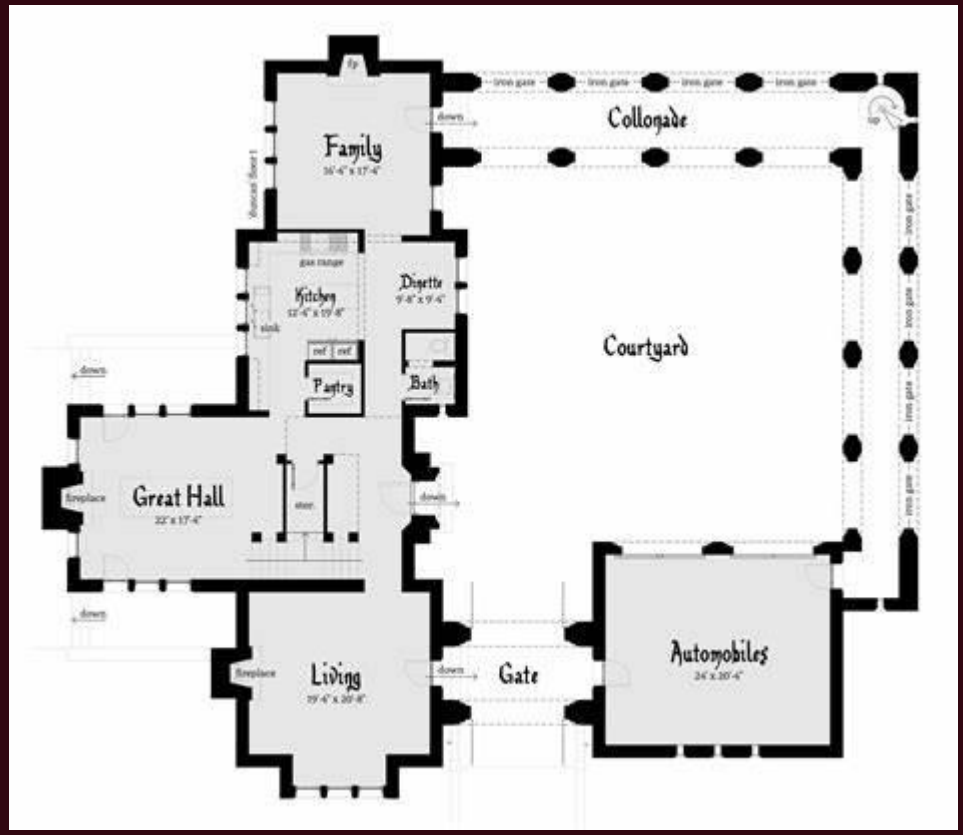
Setback

Unfortunately, working with technology comes unpredictability, in the last week of March I launched my project to find it was blank. My outliner showed all my assets however they were not visible on the viewport. I scoured google and reddit forms to no avail. This ended up being a blessing in disguise as I much prefer the next castle design I created. However, this does mean there is a lack of documentation of the first deign.



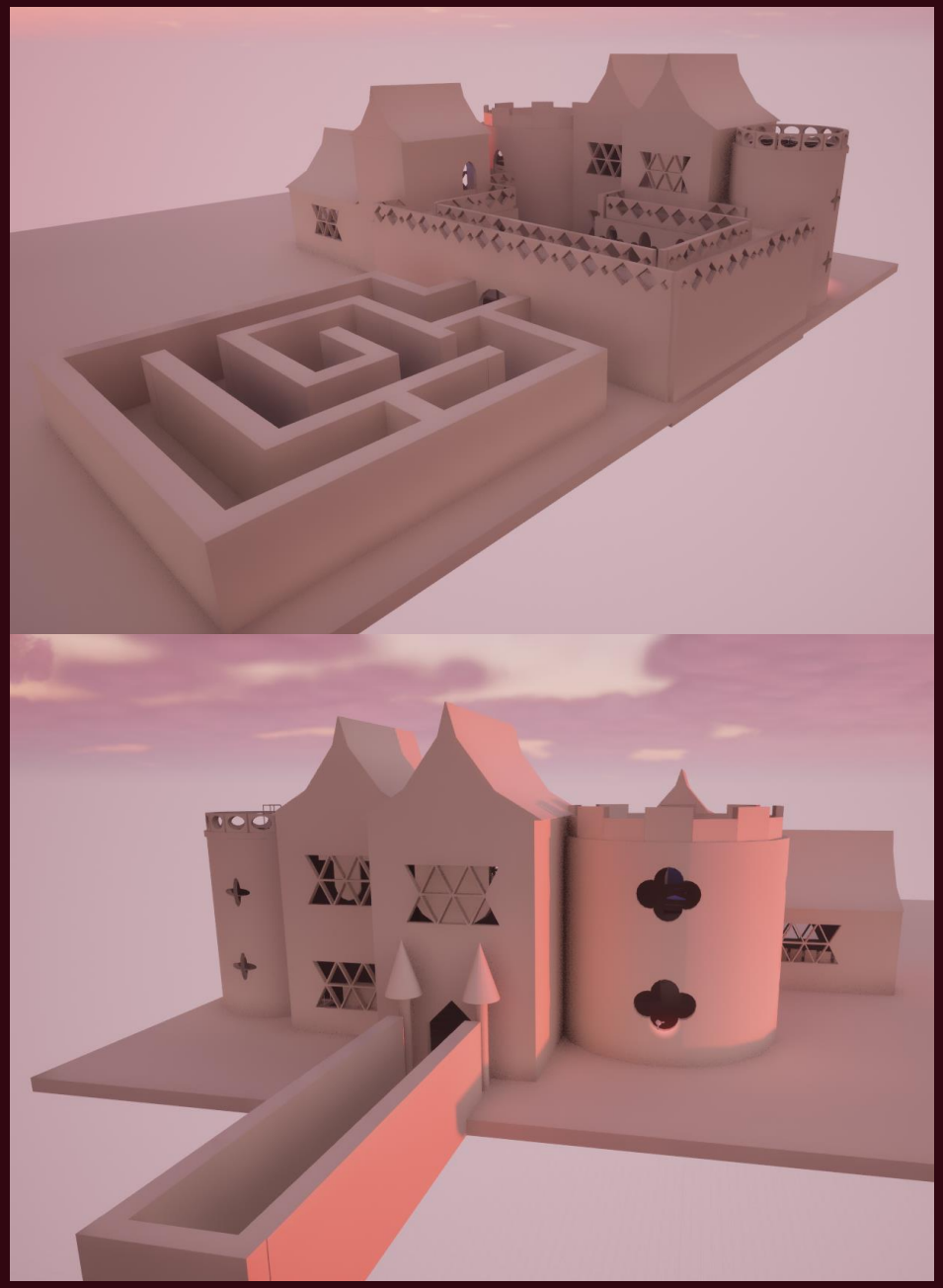
Level design – second iteration

I used this floorplan image as reference for my second version as I found it flowed better and allowed enough range for the mechanic to develop accordingly. I made small changes by adding 2 turrets on the outer corner structures, so it better resembles a castle.



Second level iteration exterior

- I made my own twist on this layout by adding turrets on the outside and creating a maze off the courtyard. This suited my mechanic much better.
- I used a basic white texture for all models to have the whole map appear more uniform.



Design







- Although art and aesthetics isn't a major concern of the project, I wanted to ensure the nature of my mechanic was communicated through visuals.
- I utilized unreal engines modelling tools to create these whimsical shapes.
- Most of the design is inspired by symbols on playing cards, much like '*Alice in Wonderland*'.

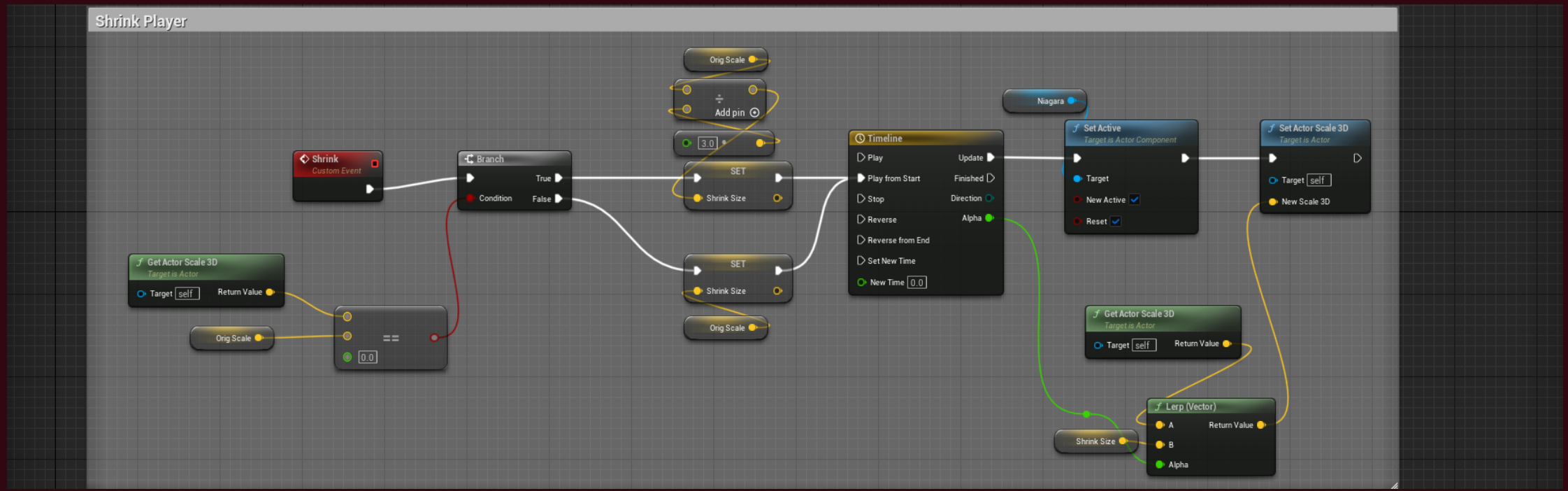




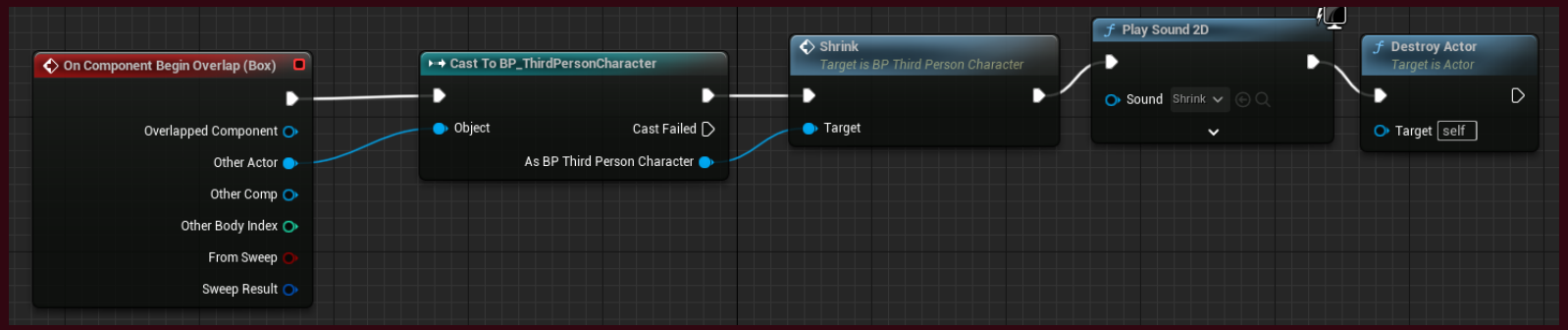


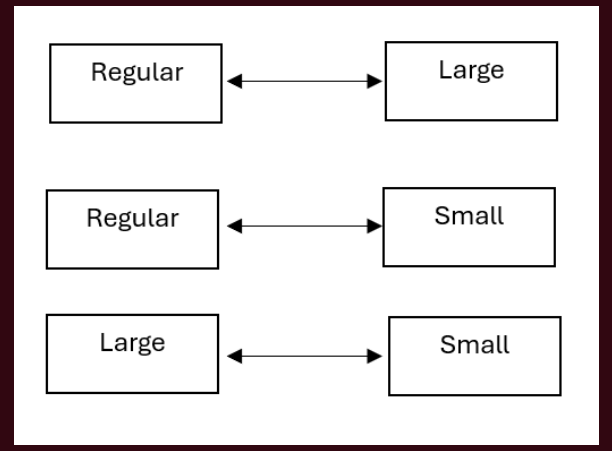
Implementation of Mechanics

I implemented my core shrink and grow mechanic in blueprints. I took the original scale vector value and divided it by the shrink scale vector and animated this size decrease through a timeline. I also trialled use of visual effects using UE's Niagara system however this would continuously crash my project, so I removed it in the end. I took this same logic, replaced divide with multiply for the player grow blueprint. I would then link these to collectables.

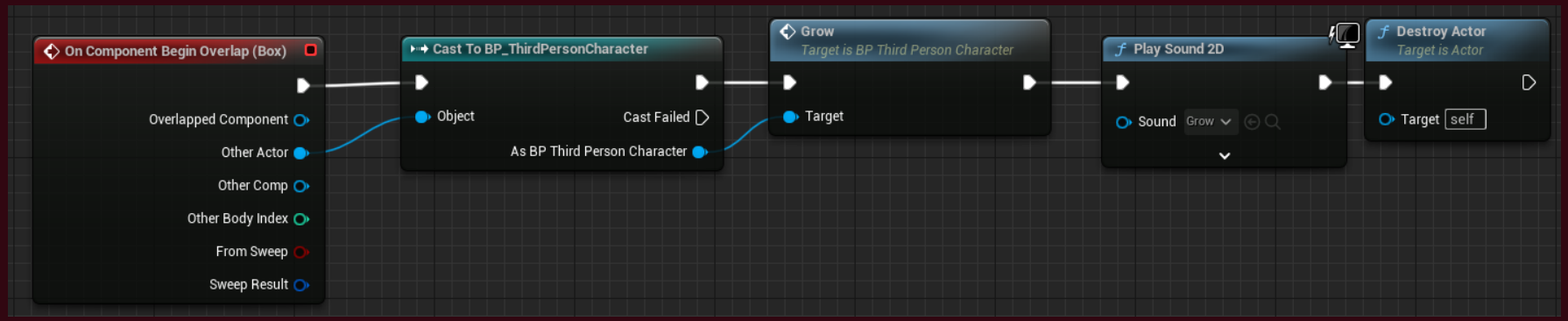


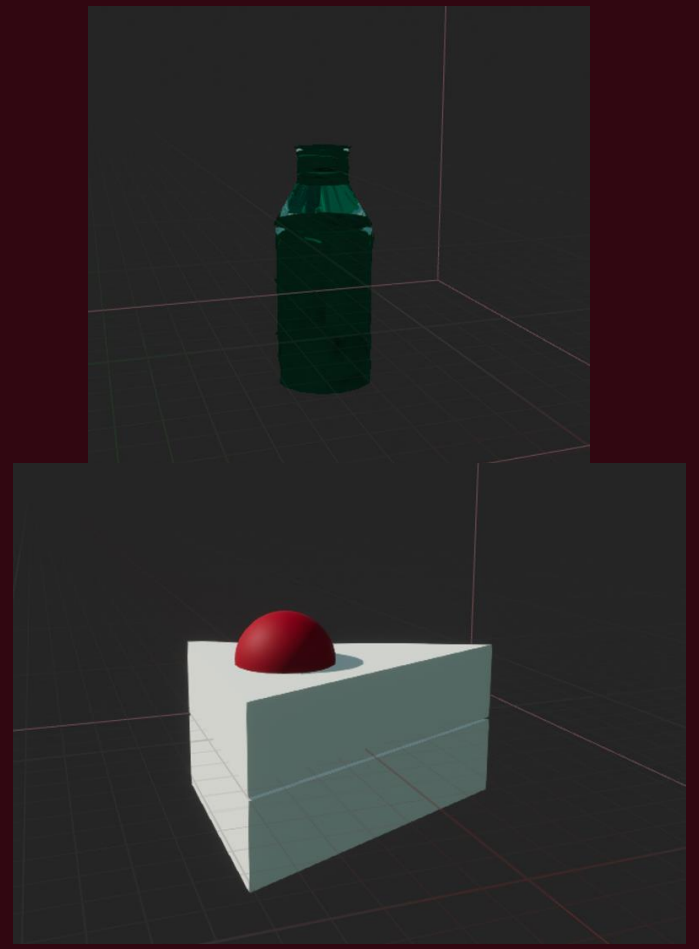
- I created a quick diagram to help me visualise which states the player is required to change from.





- I then modelled a bottle in engine and implemented the code for shrink.

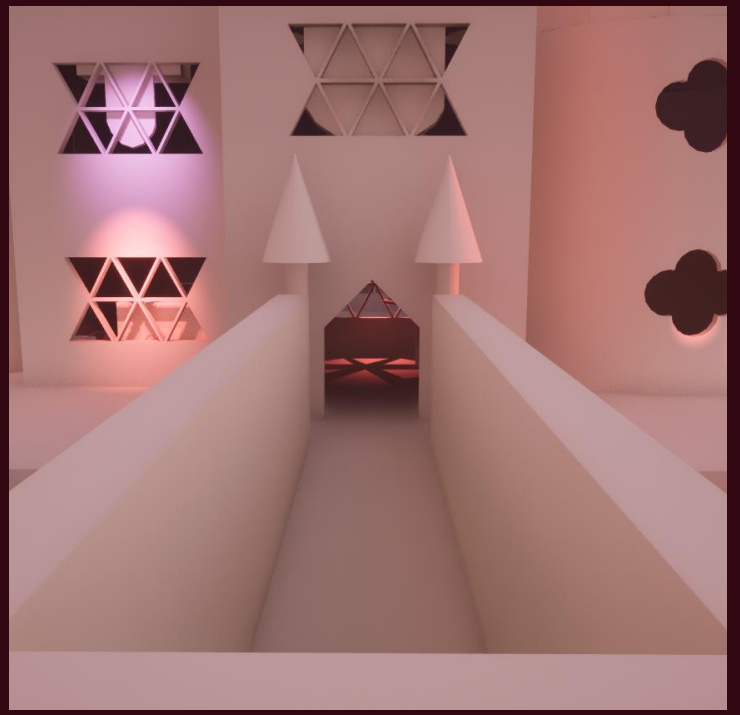




- I also modelled a cake slice in engine and implemented the code for grow.

Spawn

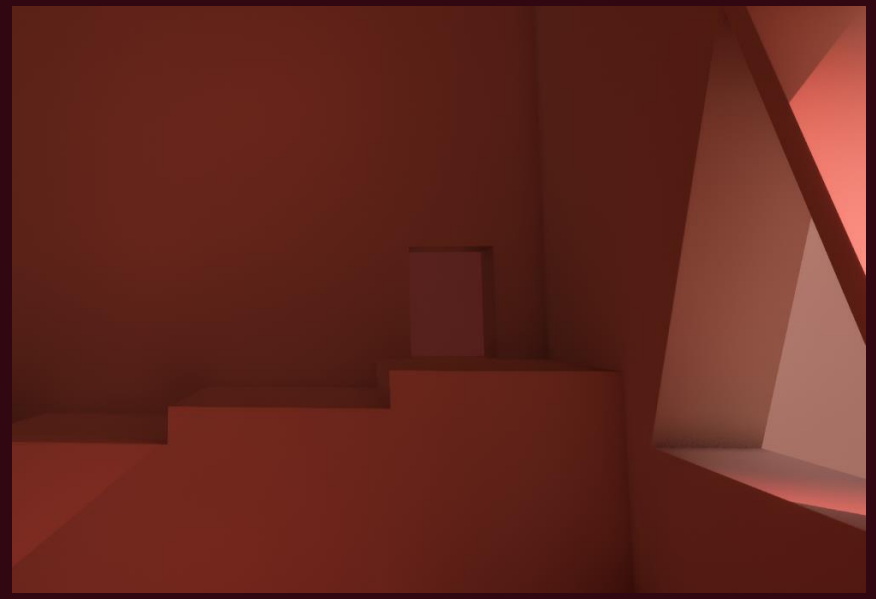
- Created a stretch lead up surrounded by “hedges”.
- Players gets their first look at the castle, builds anticipation.
- Gaps next to spikes filled with invisible walls so player cannot get out of bounds.
- First easter egg is behind the player spawn.



Entry room

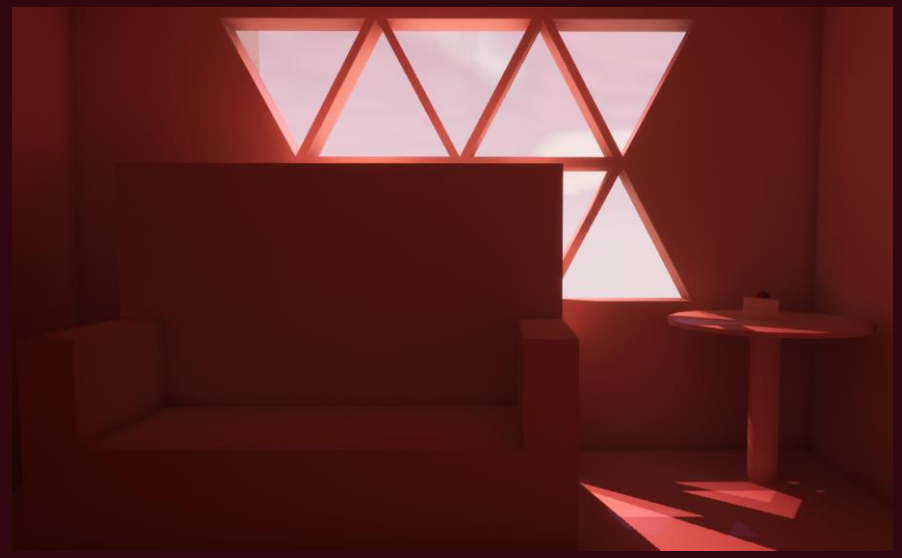
- Introduces player to mechanic.
- Sparce room diverts the players attention to the staircase where they'll find they are too large for the door.
- Their view is driven upwards to the bottle on the table.
- Lastly, they will locate the cake slice hidden in the shadows by the base of the staircase.
- They will collect the cake, then bottle, then follow through the door onto the next room.

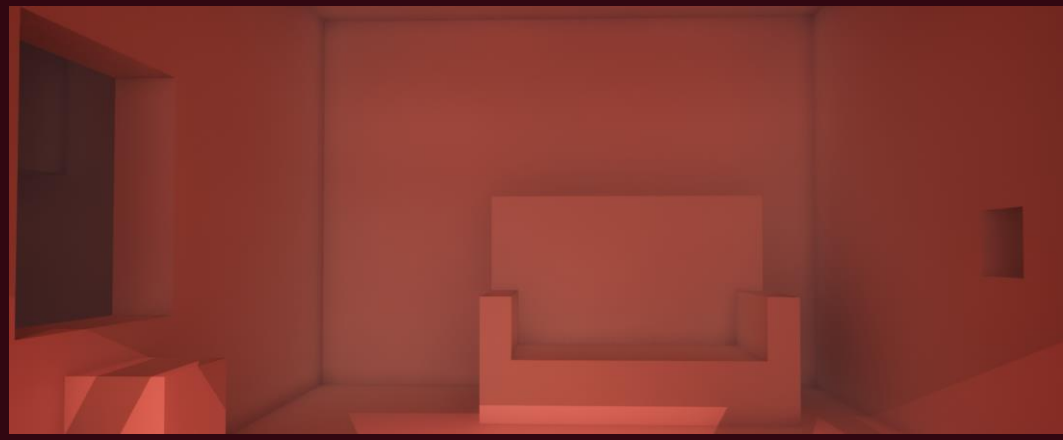




Lounge

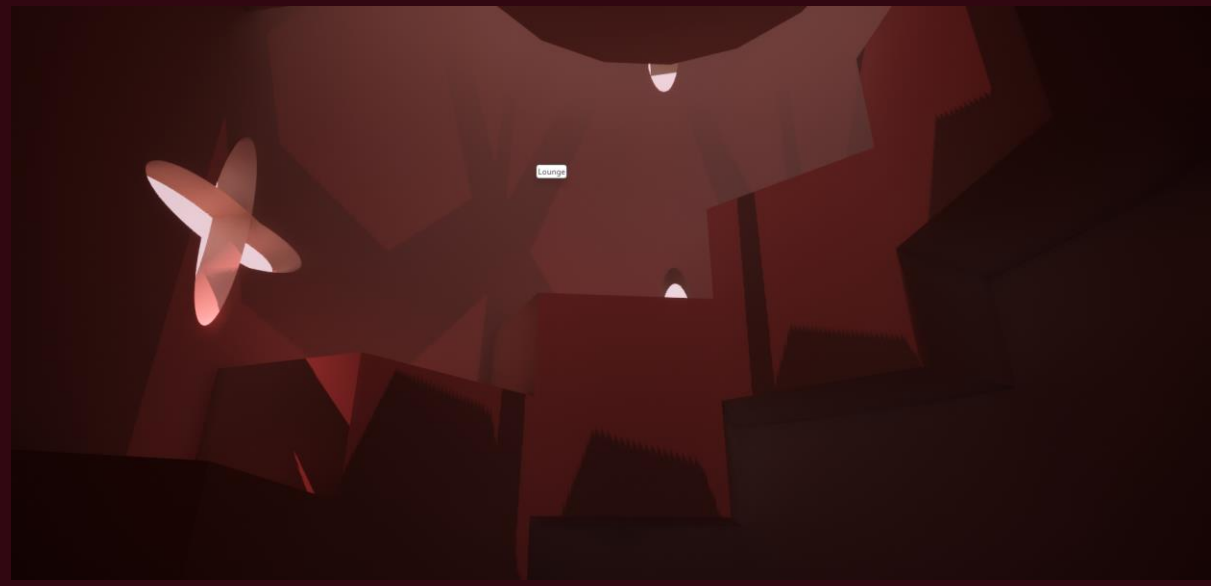
- Player will find two sofas and a table, a ledge and a normal sized door.
- The cake is simply located on the table in plain sight.
- Player may try and jump on the sofa to get through the door, if they are successful, they will be too small to progress further.
- This teaches the player that any attempts at shortcuts will not convenience them, the level should be played as intended.
- The second easter egg is located behind the window sofa.





Stairwell

- The player will progress up the spiral staircase.
- I included an extra circular ledge at the top since if the player fell to the bottom when large, they would get stuck.
- The next collectable is located at the top of the landing.



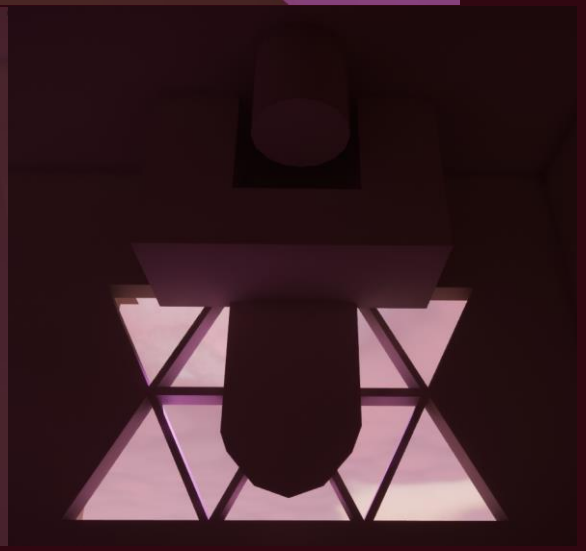


Bedroom

- Lighting changes colour to a magenta to show progression and an increase in difficulty.
- The large entryway is large, hinting to the player to seek the next collectable
- Use of lighting leads the player to the small door
- The bedroom furniture is flipped upside-down, collectable is found high up on the bedside table.







Turret

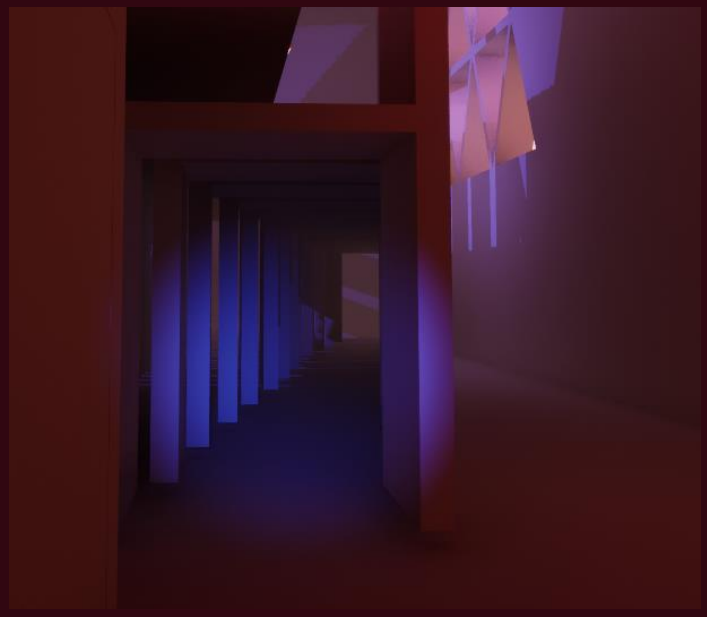
- This circular room is filled with miscellaneous boxes and a boarded-up door.
- The third easter egg can be found under a leaning box.
- The light through the door creates an arrow-like shape pointing to the hole in the floor for the player to exit through.
- Once through the hole, the player enters an empty room where again lighting will lead to the door.



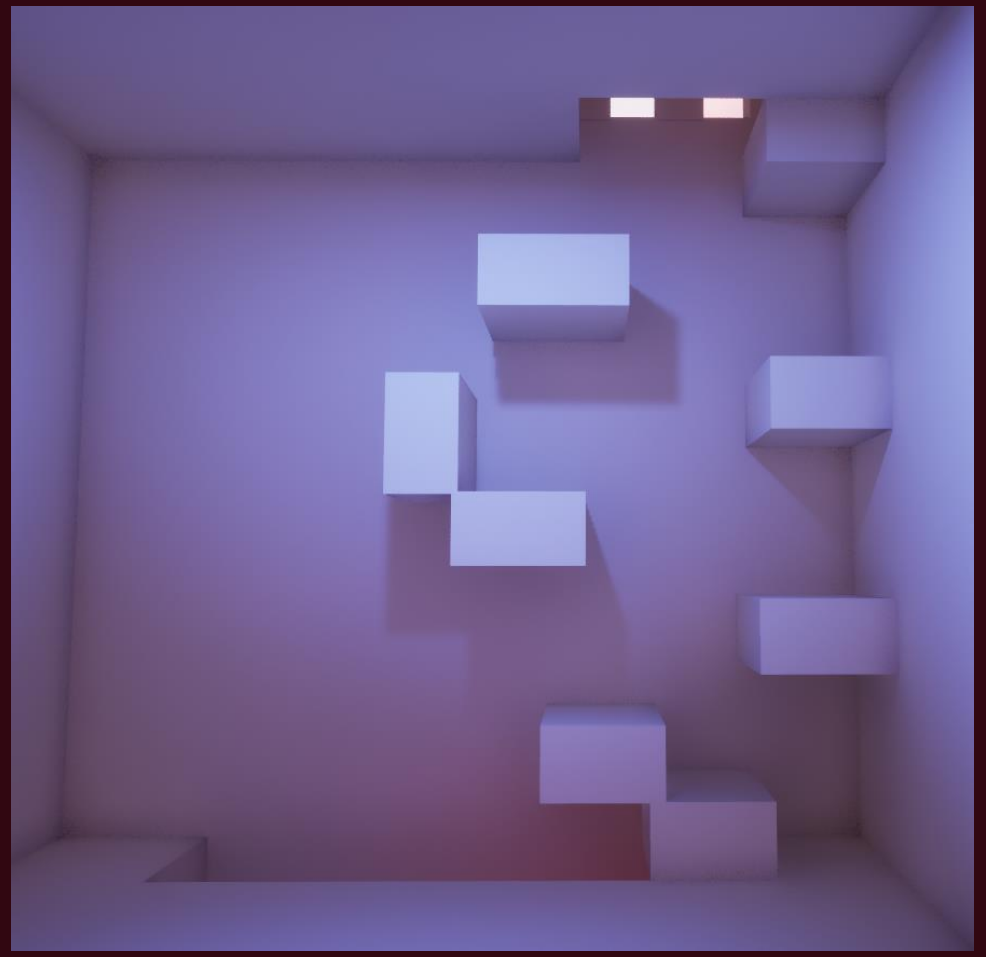


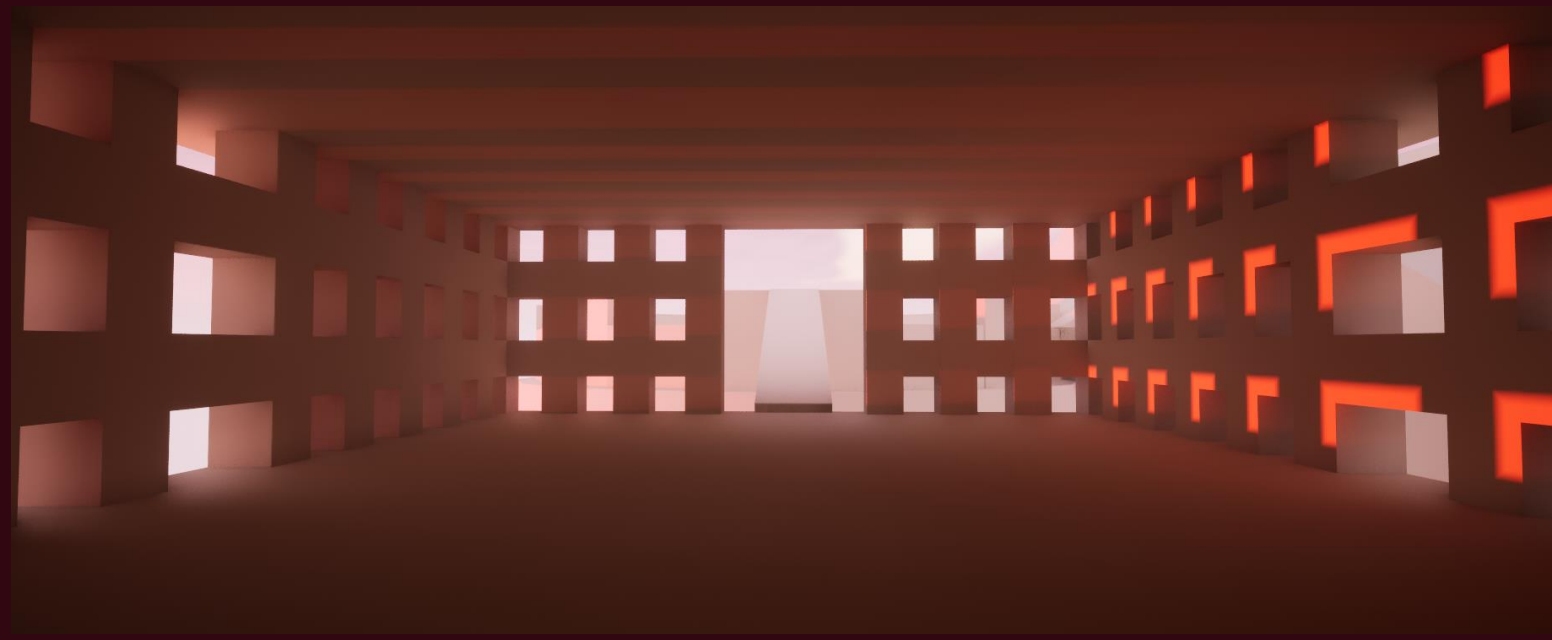
Dining Room / Tea Party

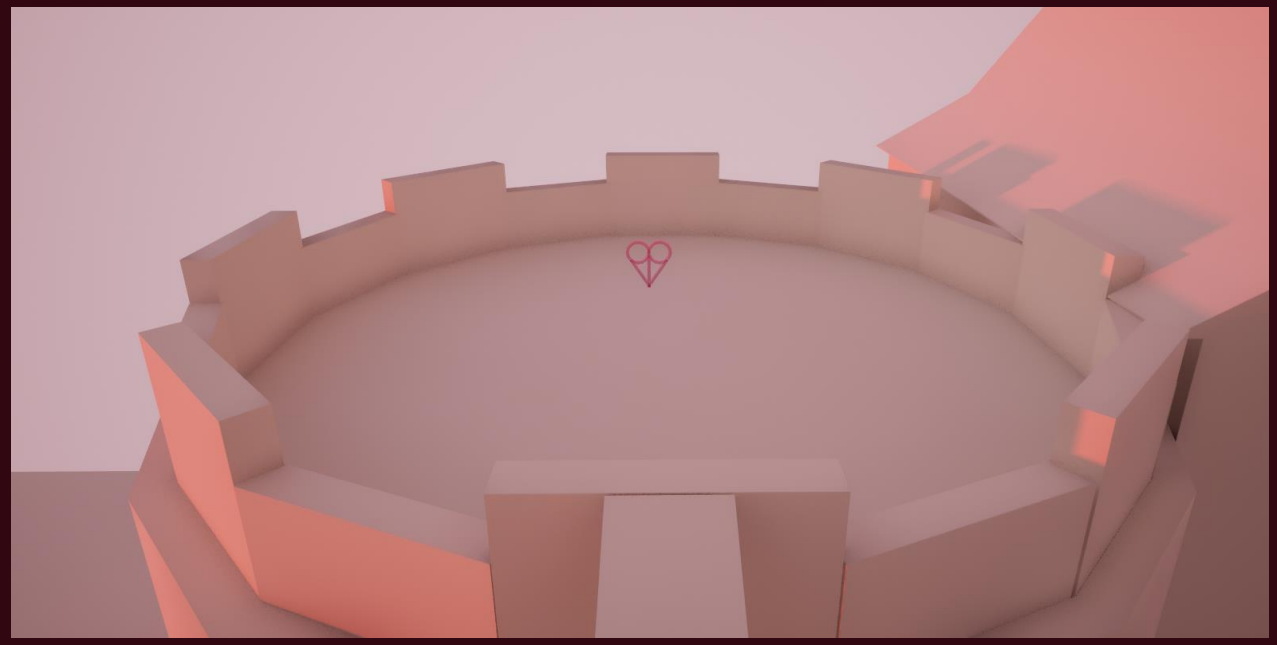
- The lighting changes to a blue colour when reaching this room to show further progression.
- The player is led through a tunnel of chair legs toward the staircase.
- They can then use the staircase to continue upwards or jump onto the table filled with ploy cakes and bottles, with 1 real cake.
- If they chose to go upstairs first, they will fit on platforms that will lead them to the 4th easter egg, they will then have to go back on themselves to become big enough to jump over the barrier.
- If they chose the table first, then they will continue straight through the barrier over the balcony.







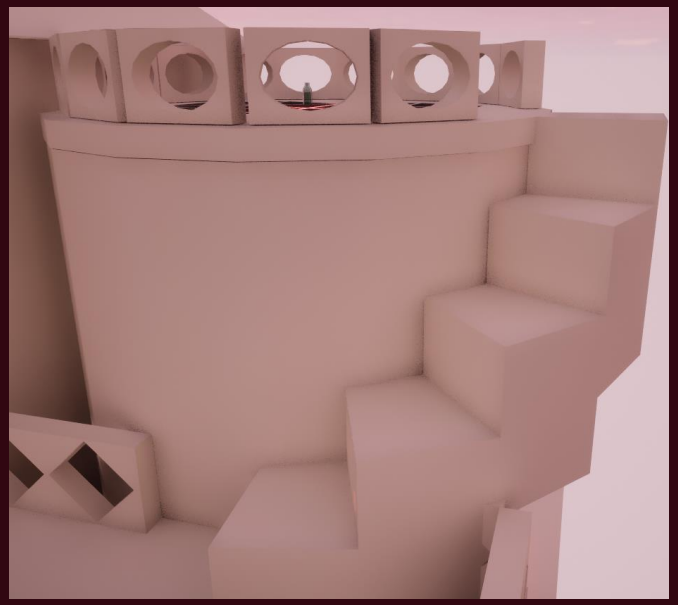


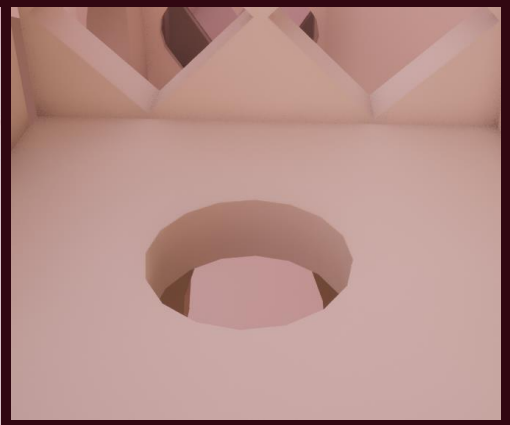


Balcony

- Initially the player will be met with a collectable cake, so they grow large, giving them a good view of their surroundings.
- They will come across a spiral staircase onto the roof of the turret where the collectable lies to make them small enough to progress through the next hole.
- I placed invisible walls around the staircase and above the fences so the player cannot jump out of bounds.

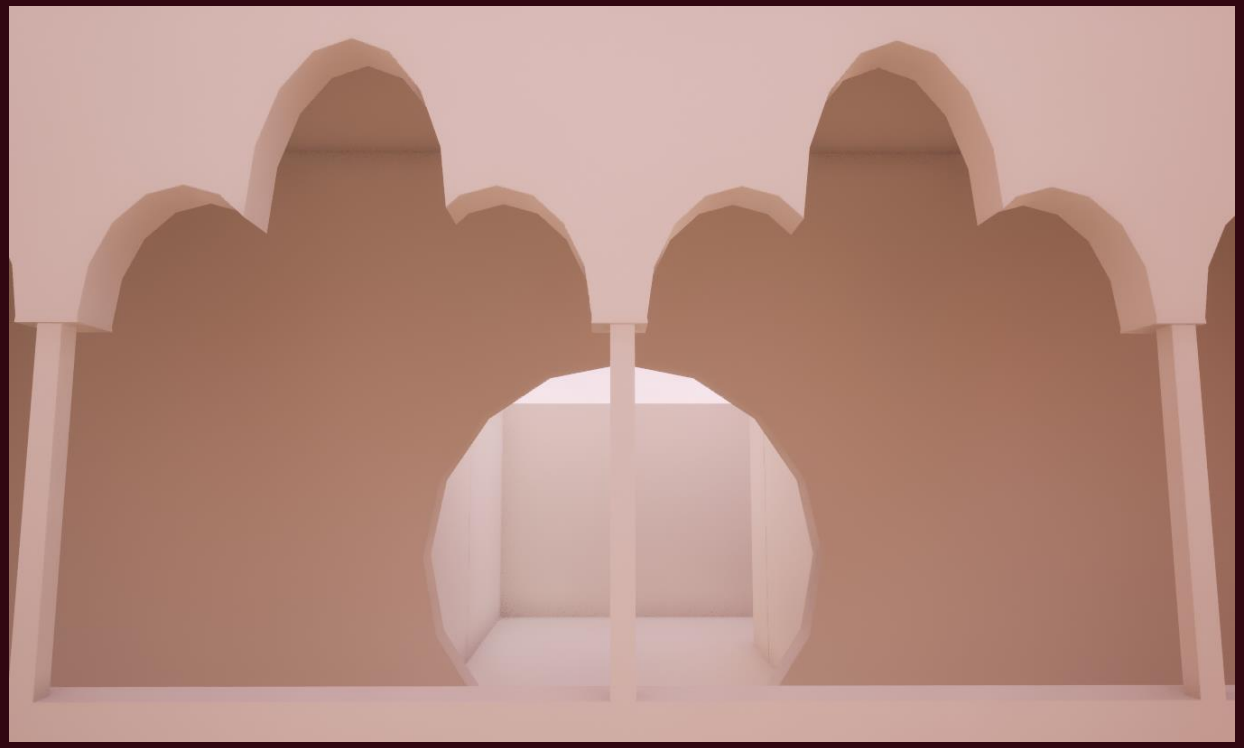






Courtyard

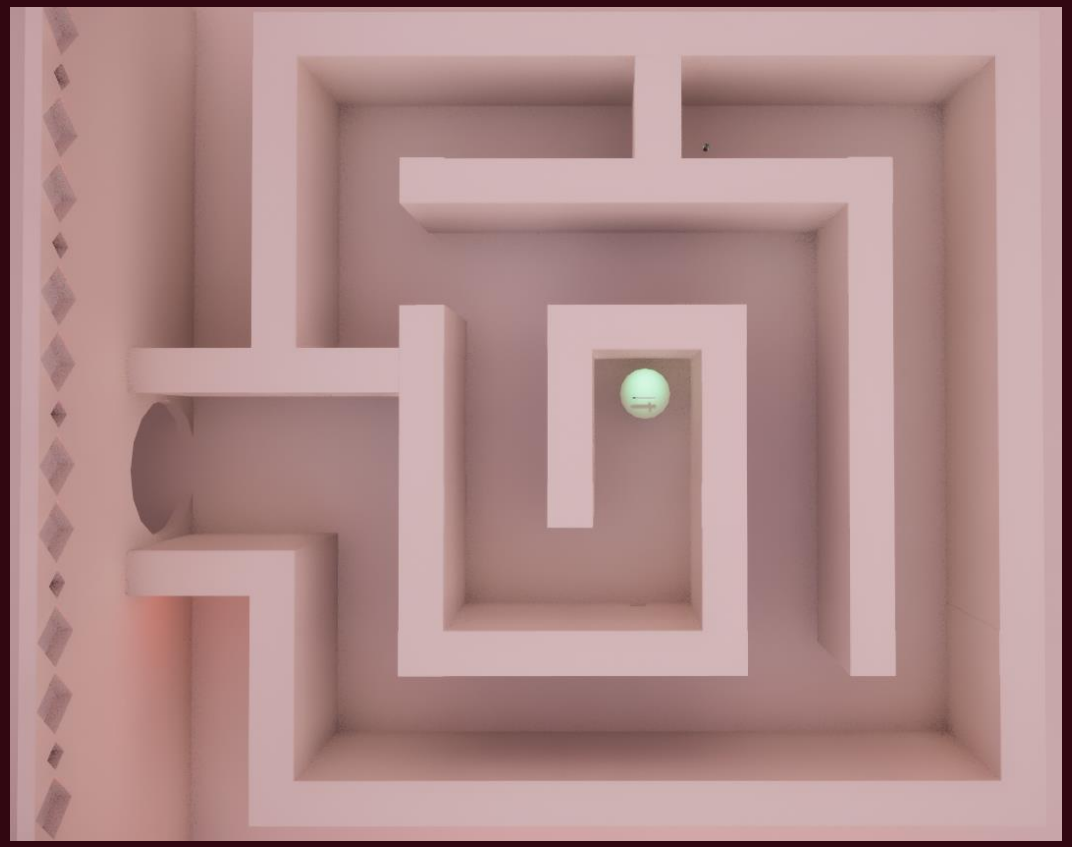
- The player will be led to the round door that takes them through to the maze.
- If they continue to go further round, they will find the fifth and final easter egg in the level.

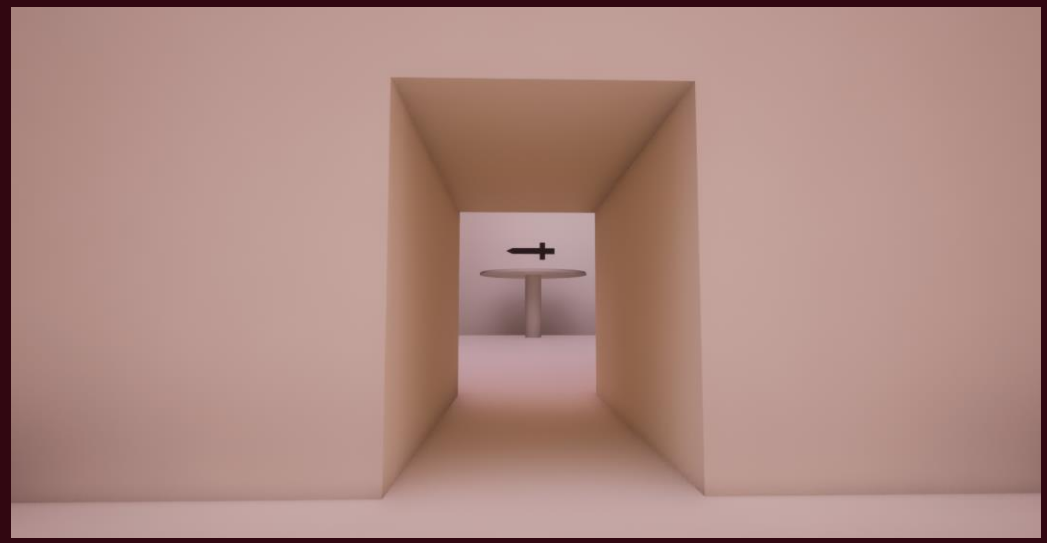




Maze

- On the second turn of the maze, the player will see a small hole in the wall, this prompts them to search for a bottle.
- If they take the second left turn, they will find the bottle and take the shortcut.
- If they take the alternate route, they will eventually reach the sword where the level will conclude.





Testing (V2)

Tester 1:

- Wasn’t completely clear of the objective
- Can get over barrier onto terrace when small – not intended.
- Wasn’t sure if they were meant to go on dining room table, didn’t find collectable.

Tester 2:

- Got thrown up by the collision on the easter egg behind the sofa.
- Struggled to jump on platforms.
- Also wasn’t sure about dining room table.

Notable:

Both testers never seen platforms to lead them onto the room above the dining room. This is okay as it is an easter egg

Both testers took the outside route of the maze to discover the collectable and went through the hole, as expected.

Both testers tried to jump on the table in the starting room.

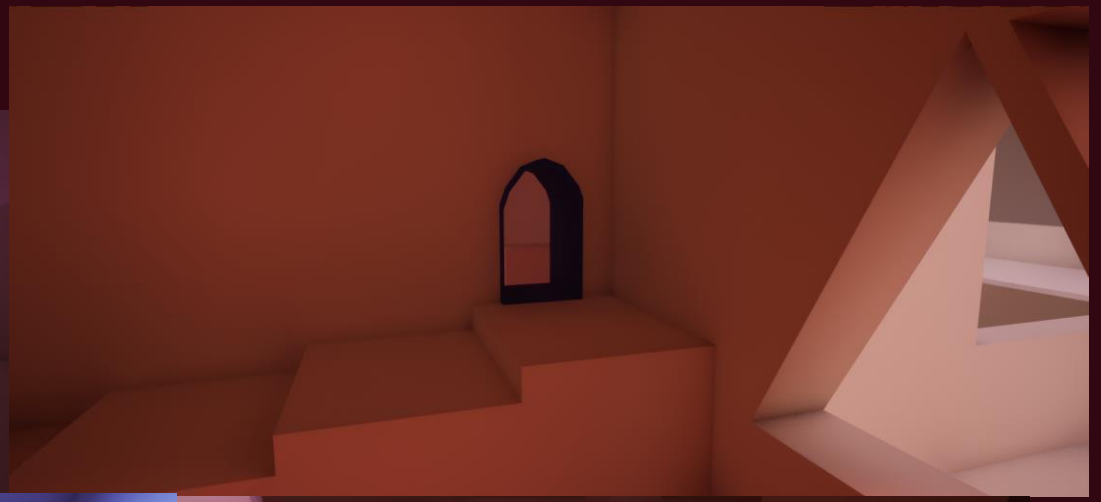
Next steps:

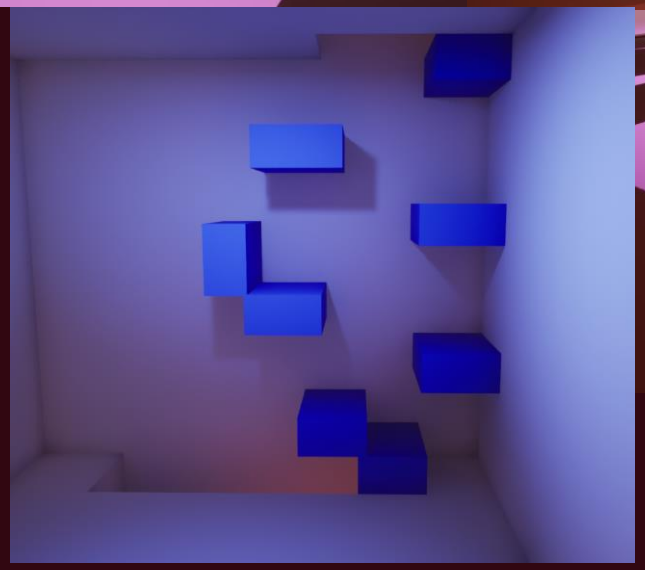
- Colour code the doors, small, medium, large.
- Keep consistency in doors.
- Make stairs wider/easier to climb I don’t want this to be a challenge for the player.
- Make sofa arms wider so easier to platform.
- Create a wider range of jump heights for each player size.
- Add more lighting.

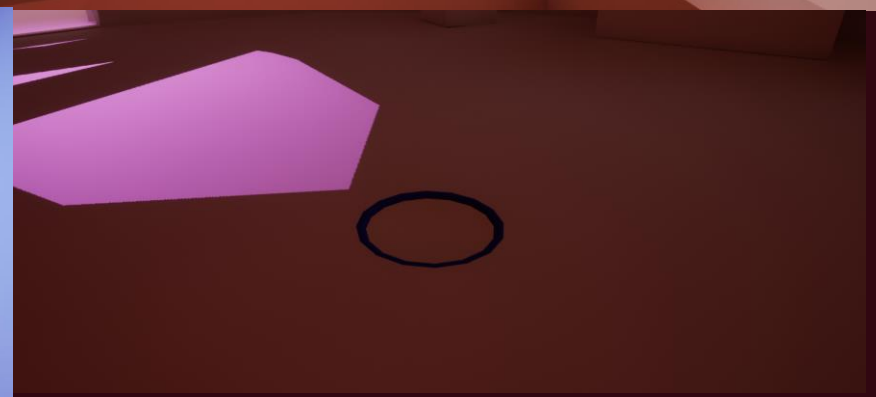
Changes I made

Since testers were unsure of the objective and were confused about where they were able to fit depending on their size, I created a consistency by colour coding areas accessible depending on size. In this example, blue represents areas accessible in shrunken size.









More examples of this, pink representing standard size.







“Establish a visual language within your level to clarify where players can and cannot go” (Rodgers, 2014, p. 246).

To persist consistency and to encourage association within the player. Any area were blocked off would have the same appearing “planks” blocking them. This ensured that when the player would come across these, they'd know that area was inaccessible.





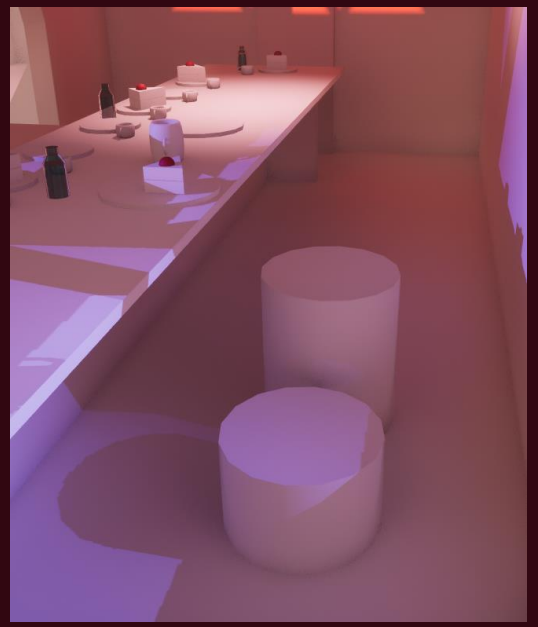


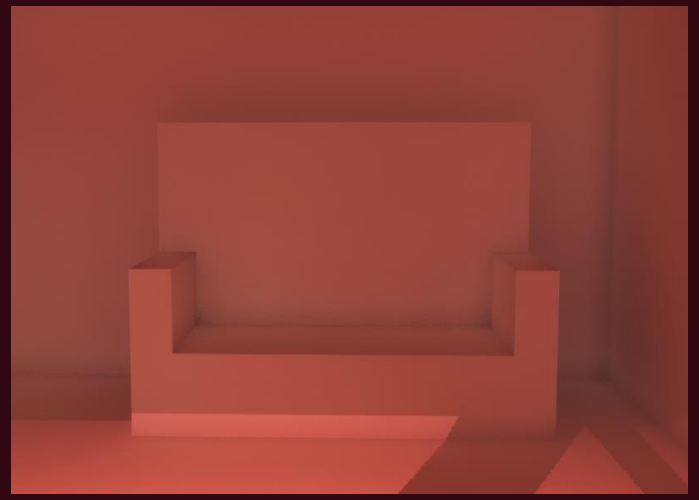


Other changes I made were:

- Creating larger stairs in the exterior balcony so that it was clear to the player they must be large to walk up them.
- I added to stepping stools that leads the player straight on top of the grow collectable so that they would not miss it.
- I reiterated the sofa design so the player would have an easier time platforming off the arm.
- I also made the table in the entry room taller and situated it in the corner so players wouldn’t try and jump on it.





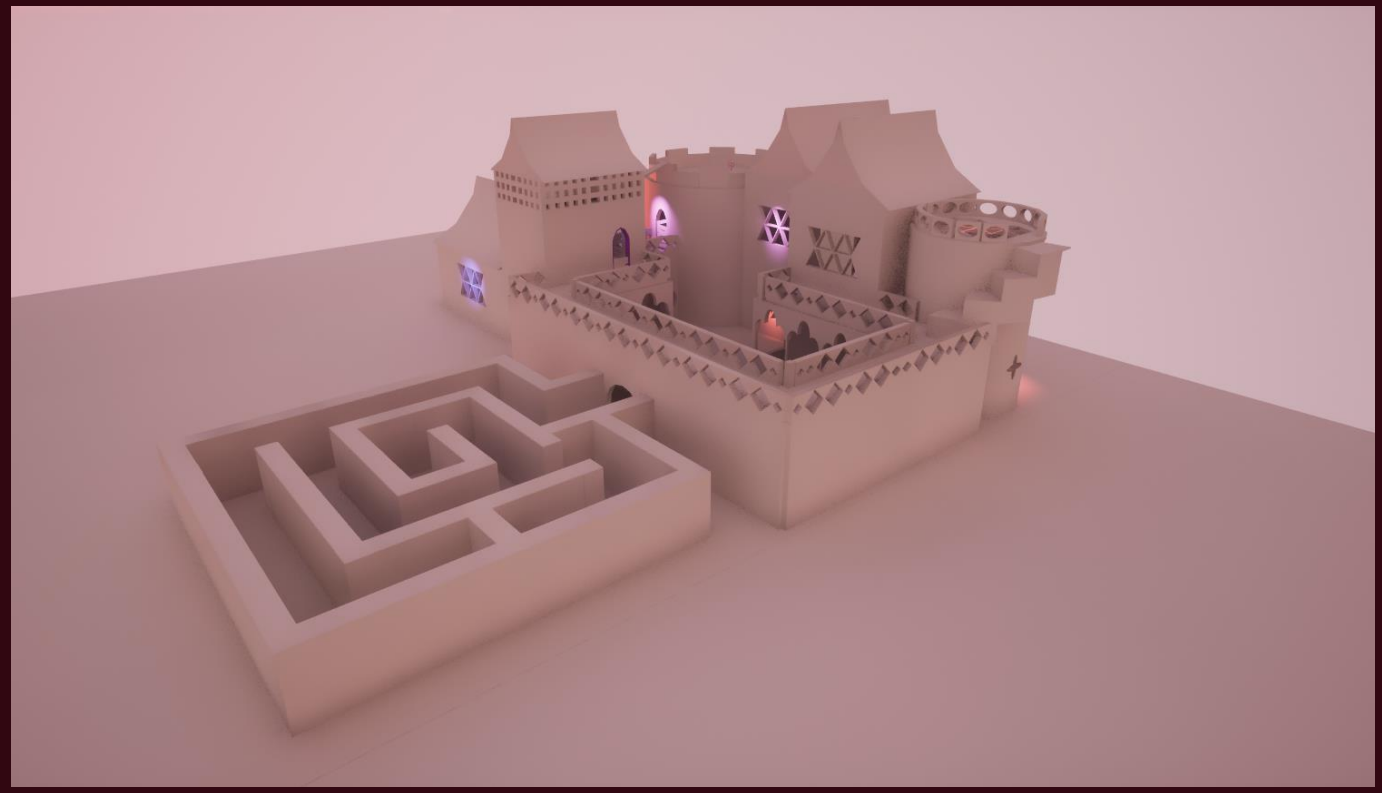




before

after

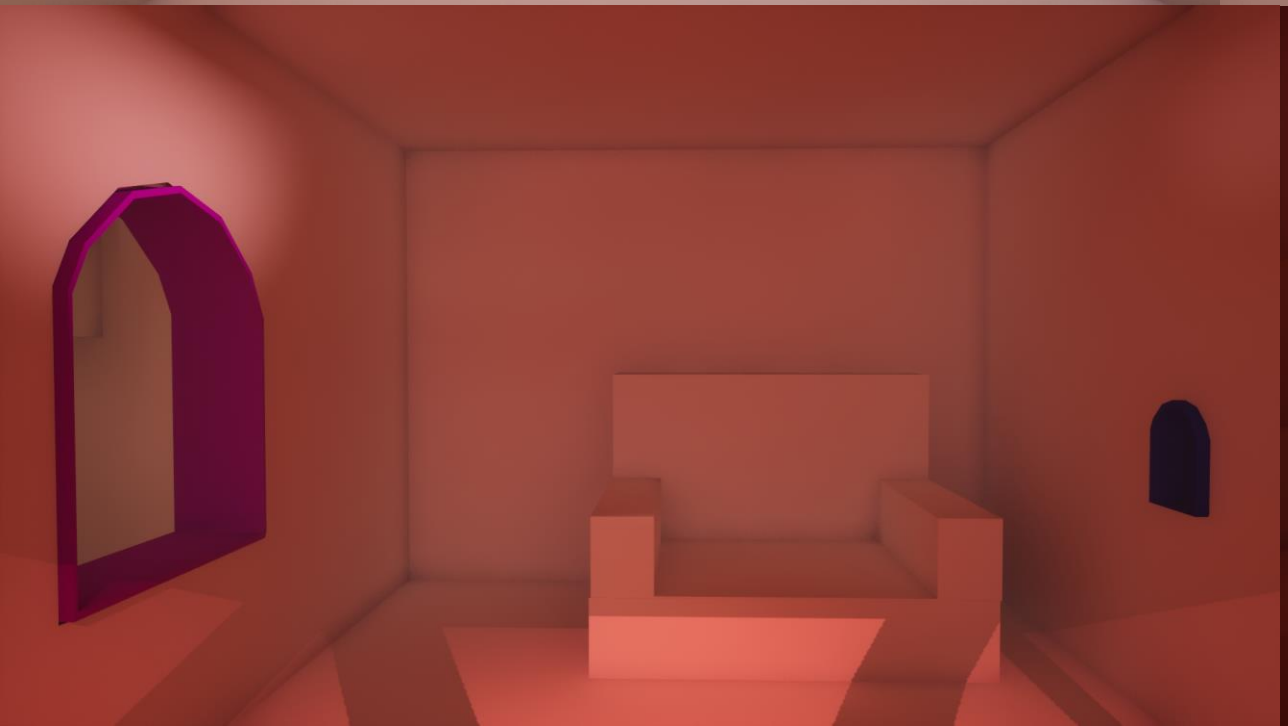
General exterior updates included an extra area above the dining room to allow the player to access the opposite turret to find an easter egg.



Final Prototype



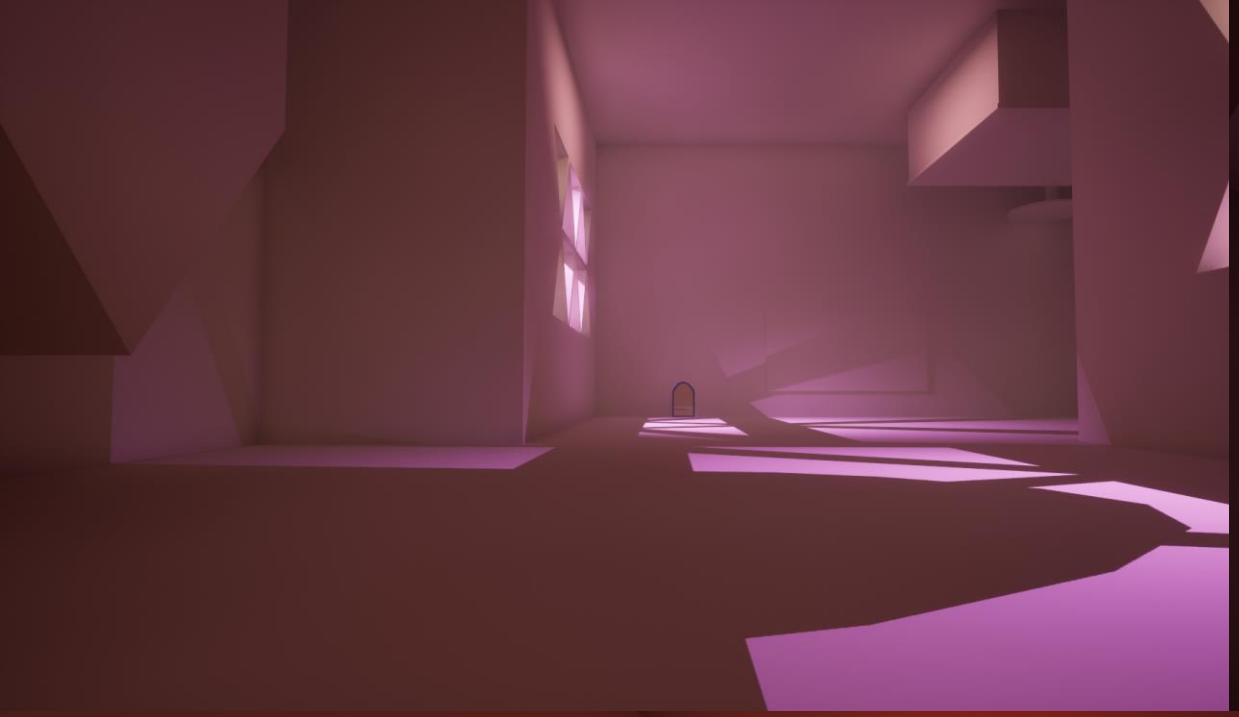

Drink me

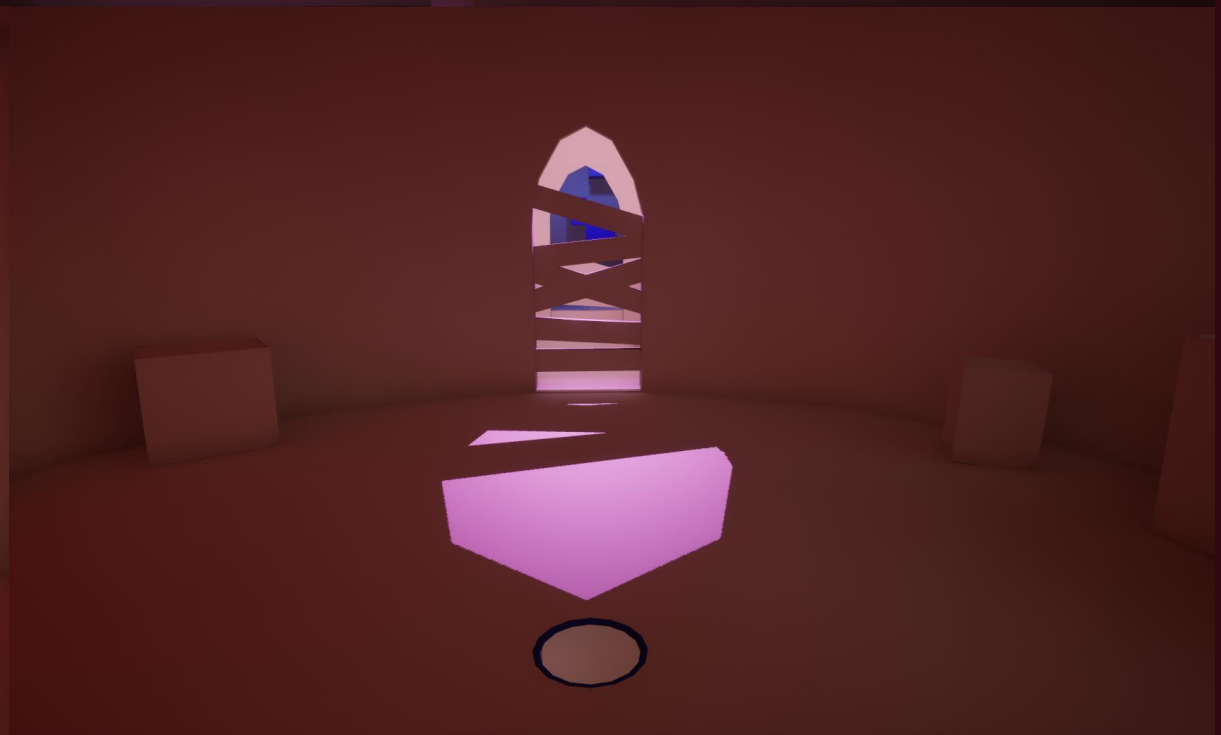


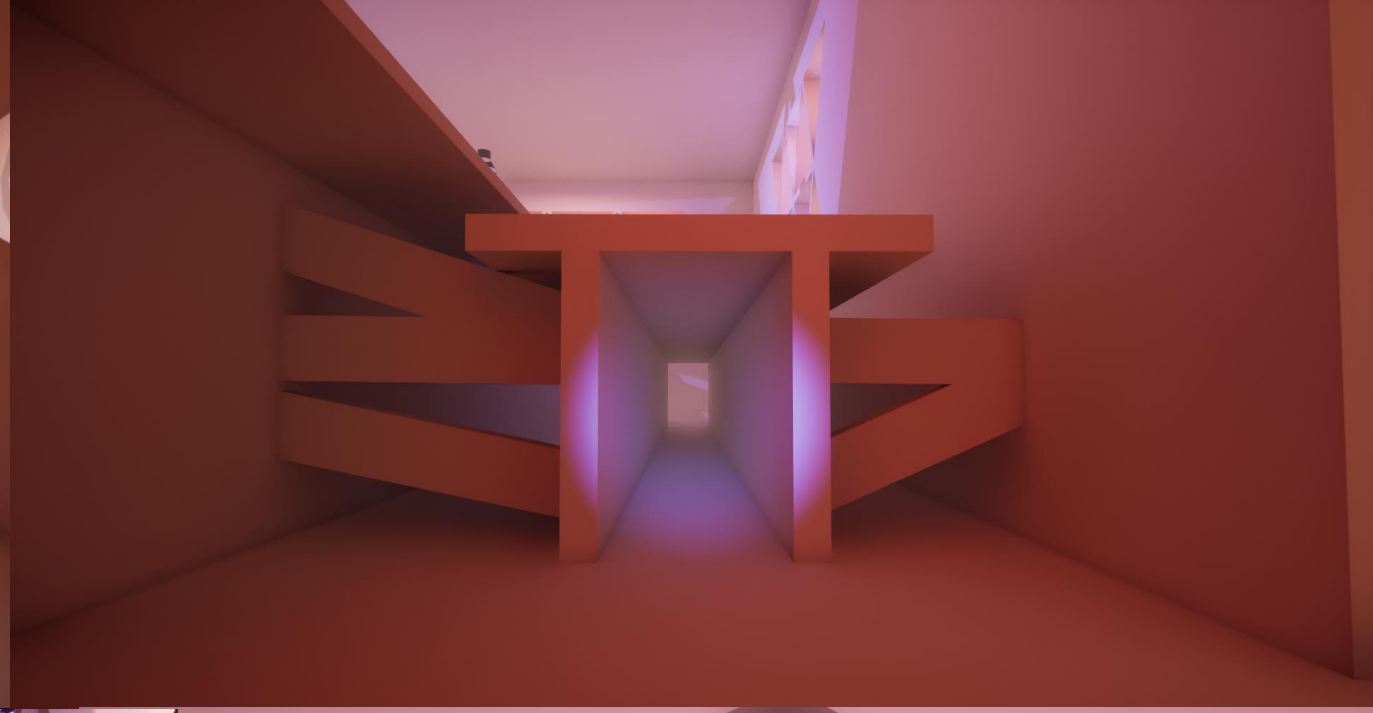


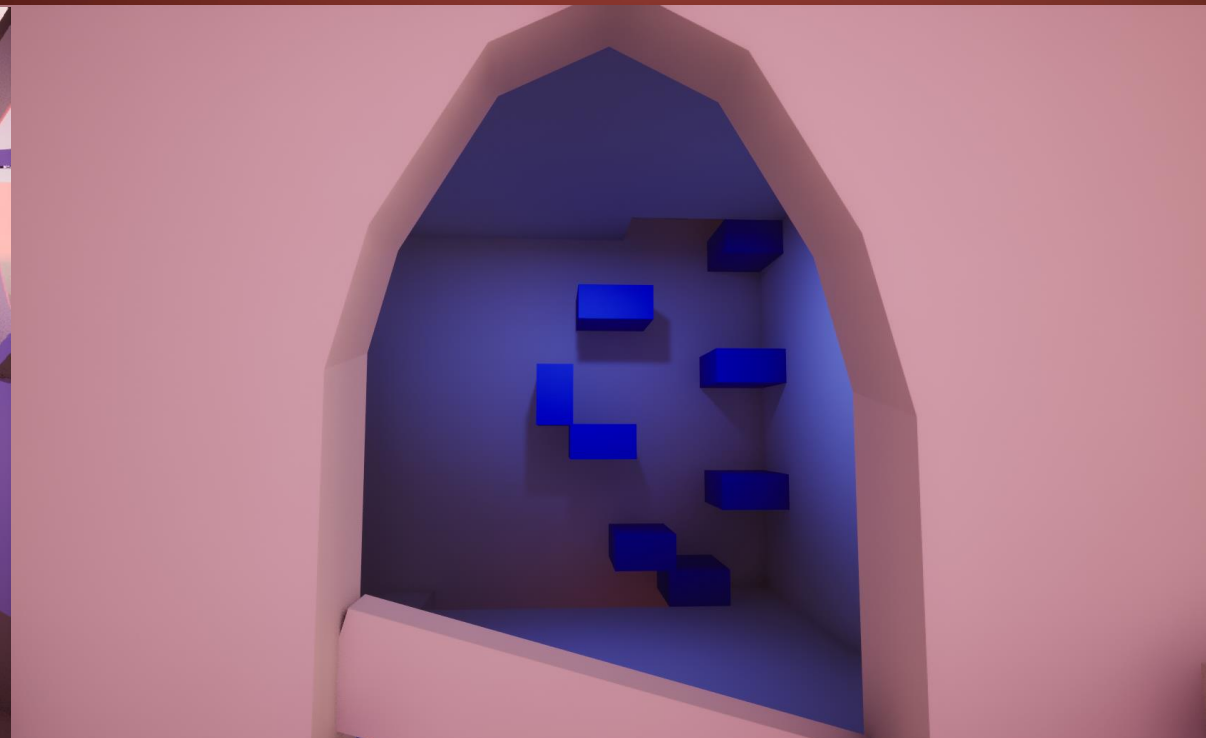


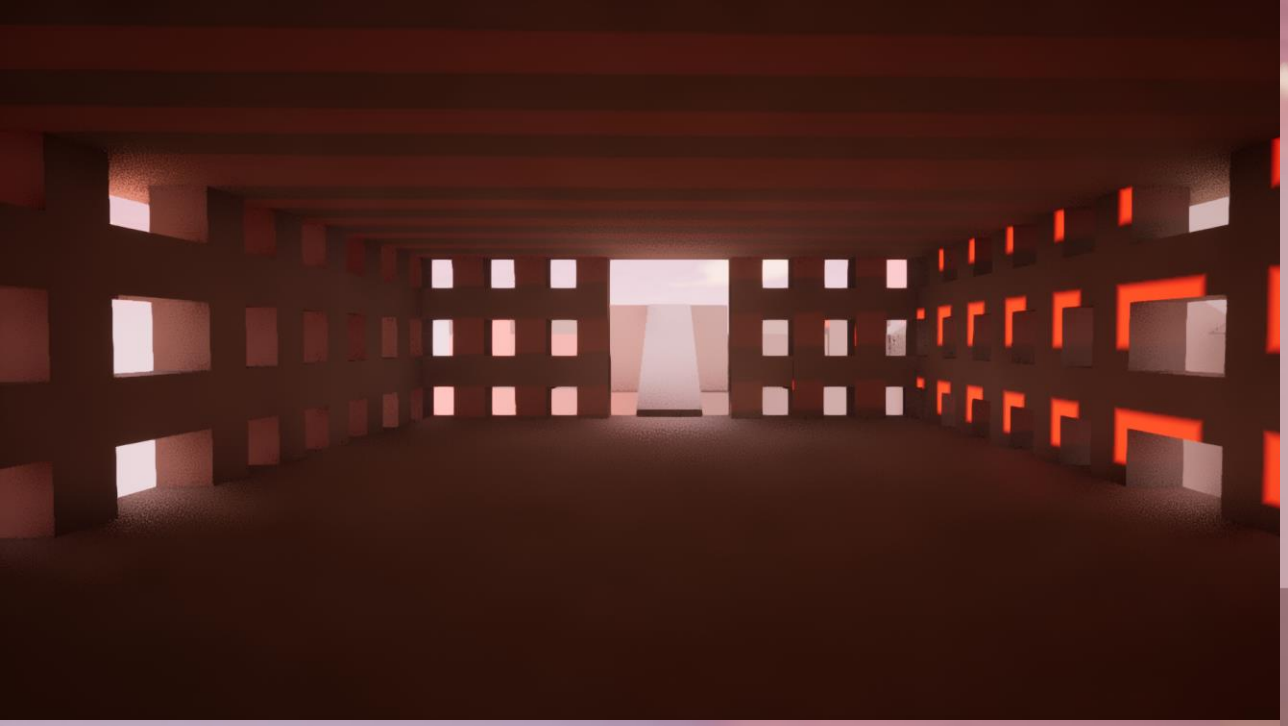


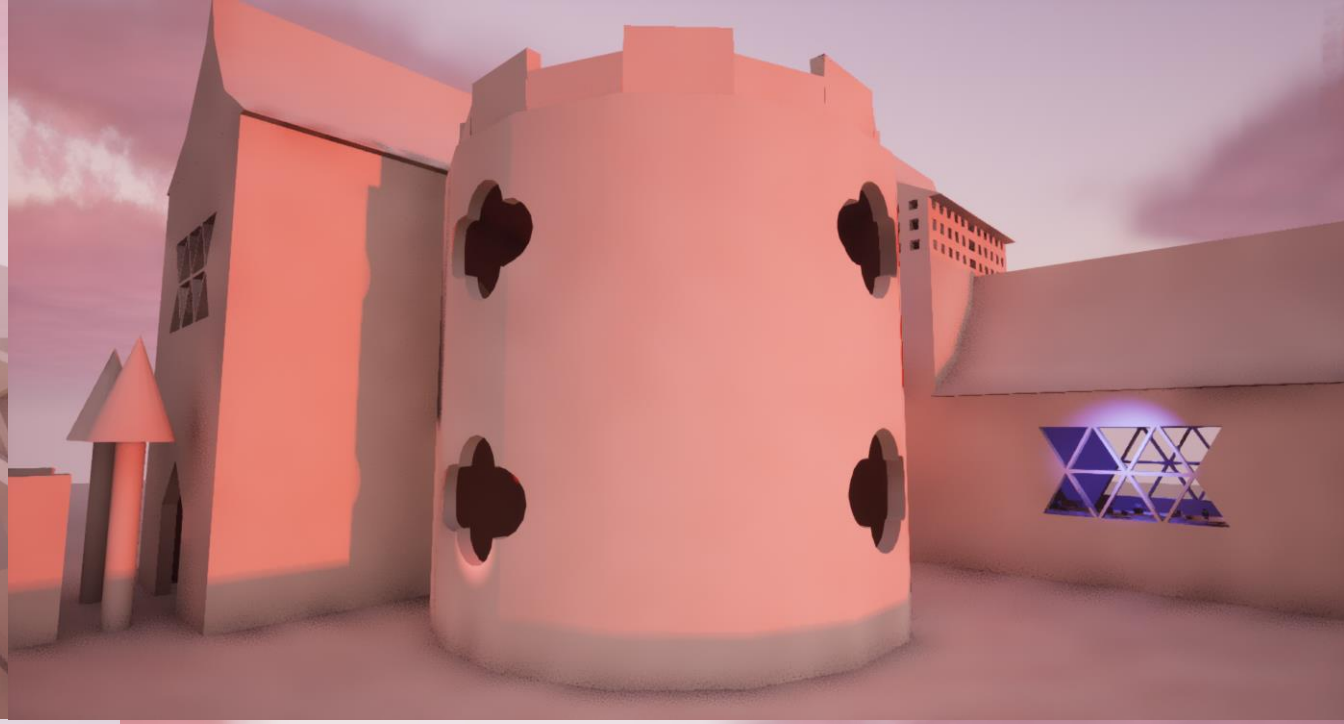








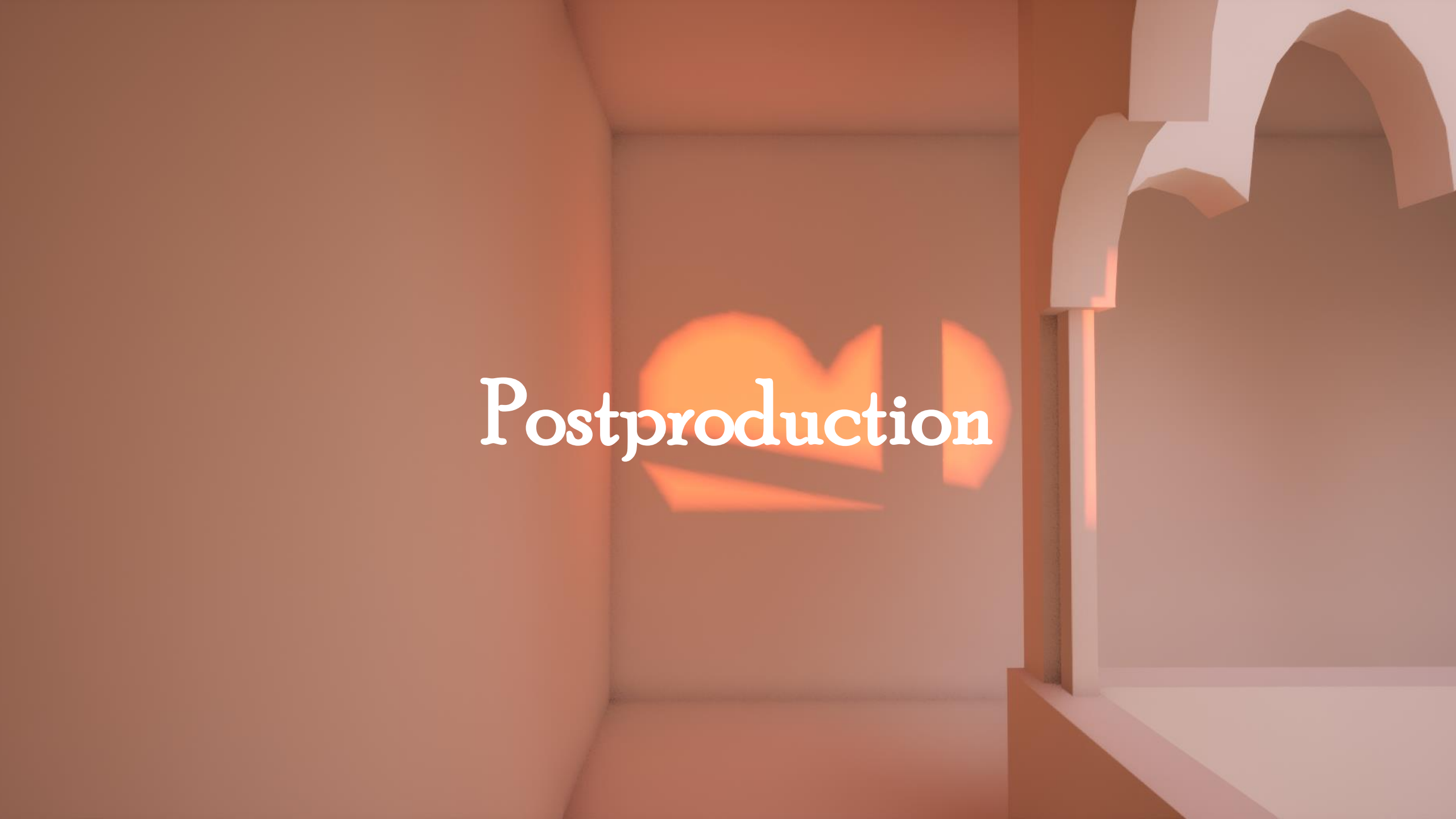










Postproduction

Critical Reflection

Now reflecting on the project, I generally feel I have met the aims in which I set myself at the beginning of the semester. As much as I would've benefitted from focussing this project strictly learning C++, shifting my focus to level design was much more suited for the workload since also juggling the mass of DES310. Level design is very interesting and the material on elements to learn is endless. There was a lot that went well in the project and a few things I would like to work on.

Starting with the negatives, a self-goal I set for myself was time management as this is certainly a gap in my skills diamond. I made use of the gantt chart I scheduled for myself in the first week, even giving myself extra time for any hiccups. In the beginning weeks I was sticking to this schedule well however when I reached the implementation phase I veered off schedule as the middle of the semester was a busy time. Losing the project didn't help matters either. However, since I had accounted for this within my gantt chart, the disruption wasn't huge fallout of the project.

I had also faced issues with scripting not giving the results that I had intended. This took up more time than planned since most of my implementation time was dedicated to grey boxing and polishing. The issue I had faced was the player being stuck in a low gravity when transitioning from small to their original size. In attempts to create a differentiation between movement, especially jump height when in the variable size states, I assigned each size their own gravity scale, small having a scale of 2, standard having a scale of 1.4 and large being 1. I understand these values may seem random but it's what felt best in gameplay. Eventually I would fix this issue after dedicating days to it. I wouldn't consider this as wated time as it further expanded my problem-solving skills in blueprints scripting.

The skills I learnt during development of this project will aid me as begin my professional game design journey. This semester of learning is the biggest improvement I've seen in myself yet. Despite not sticking strictly to my schedule, my time management and motivation has improved by masses not only due to this project, but other projects too during these 4-5 months.

The highlight of this project would be learning about player psychology as I find it so fascinating, and it is such a powerful tool when designing. In future I would like to further develop on environmental storytelling as this is something I'd touched on slightly in this project but not enough as I would've wanted as I find it very interesting. During the break I intend to learn C++ scripting language as this project was originally planned. This will be of great benefit to me as it is industry standard as found during research of job requirements.

Finally, a personal goal I set for myself at the begging of this project was simply to have fun. There's no point in developing an idea or discipline that I don't feel passionately about especially if it's what will lead to my career. I can genuinely say I have enjoyed developing this project on my own terms.

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